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ChemSkin DB 2.0: a comprehensive reference chemical database for skin corrosion and irritation

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Abstract

We previously developed ChemSkin DB listing 100 reference chemicals to aid the development of new approach methodologies (NAMs) for skin irritation. However, ChemSkin DB is insufficient for NAMs addressing the full spectrum of skin hazards, i.e., skin corrosion and irritation. To address this gap, ChemSkin DB was expanded with 134 reference chemicals to ChemSkin DB 2.0 as a harmonized reference database aligning skin corrosion and irritation data with UN GHS criteria. A total of 234 unique substances were consolidated and assigned to reference categories under a precautionary decision framework. ChemSkin DB 2.0 contains normalized in vivo metrics and model-specific in vitro viability values obtained with OECD TG 431 and 439 In Vitro Skin Corrosion and Irritation Tests: Reconstructed Human Epidermis (RhE) Test Methods, ensuring transparency and regulatory traceability. Especially, performance evaluation demonstrated substantial agreement across OECD TG 431 RhE models for corrosion identification. In contrast, OECD TG 439 irritation RhE models showed moderate to substantial agreement, with variability in specificity linked to the single-threshold structure of irritation classification. Chemical category stratification further revealed chemical class-dependent performance patterns across both endpoints, highlighting the influence of mechanistic diversity on model reliability. Collectively, ChemSkin DB 2.0 provides a severity-stratified, regulatory-aligned reference resource that supports transparent model development and evaluation, and improved consistency in skin hazard assessment.

Keywords Skin corrosion · Skin irritation · Severity classification · OECD TG 431 · OECD TG 439 · Reference database · Reconstructed human epidermis (RhE)

Introduction

Skin irritation and skin corrosion are critical local toxicity endpoints in the safety assessment of industrial chemicals, cosmetics, pharmaceuticals, and consumer products (Rana et al. 2025). These endpoints represent distinct biological outcomes: irritation generally reflects reversible inflammatory responses, whereas corrosion involves irreversible tissue destruction. Both hazards play central roles in regulatory classification and hazard communication under international chemical safety frameworks.

Historically, classification has relied on in vivo data generated using the rabbit Draize test (OECD Test Guideline 404 (OECD 2015)). Although this assay has provided the foundational basis for regulatory decision-making, it has also been associated with ethical concerns and limitations in predicting human responses. Consequently, substantial efforts have been directed toward the development of alternative testing strategies, including in vitro models and human-based testing approaches, aimed at reducing reliance on animal experiments (Robinson et al. 2000). Ethical concerns, legislative restrictions on animal testing (notably in the cosmetics sector), and scientific limitations in extrapolating animal responses to humans have accelerated the transition toward New Approach Methodologies (NAMs),

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including validated in vitro systems (National Research Council 2007; Sheng et al. 2025).

Regulatory frameworks distinguish corrosion and irritation using fundamentally different decision principles. As shown in Fig. 1, corrosion (Category 1) is defined by evidence of irreversible tissue destruction and is subcategorized (1A, 1B, 1C) according to time-to-effect criteria rather than numerical scoring. In contrast, irritation categories are determined based on quantitative mean erythema and/or edema scores assessed over 24–72 h. The United Nations Globally Harmonized System (UN GHS) defines a structured severity continuum spanning No Category, Category 3 (mild irritation), Category 2 (irritation), and Category 1 subcategories (1A, 1B, 1C) (United Nations 2025). While EU CLP adopts the corrosion subcategorization scheme aligned with UN GHS, it does not implement the optional Category 3 for mild irritation. Furthermore, historical systems such as the EU Dangerous Substances Directive (DSD) differ in terminology and classification mapping, creating a complex regulatory landscape (Luechtefeld et al. 2016).

These regulatory variations highlight the necessity for harmonized reference resources spanning the full corrosion–irritation severity gradient (Hartung 2016). However, existing curated datasets frequently focus primarily on irritation and often rely on partially overlapping or method-specific chemical sets. Potent corrosive substances and explicit 1A/1B/1C stratification are often underrepresented, and cross-regulatory mapping remains inconsistently documented.

Validated in vitro approaches have been established to support non-animal hazard identification, including reconstructed human epidermis (RhE)-based assays described in OECD TG 439 (OECD 2025b) for irritation and OECD TG 431 (OECD 2025a) for corrosion. These advanced three-dimensional tissue models represent key components of modern non-animal safety testing strategies and have been extensively reviewed in the context of emerging NAM platforms (Alépée et al. 2014; Ezendam et al. 2016). The transition toward Integrated Testing Strategies (ITS) is essential for modern safety assessment, as these strategies enable the integration of multiple data sources to improve predictive reliability (Rovida et al. 2015). However, these methods are typically optimized for specific decision nodes and evaluated against limited reference sets, constraining cross-method comparability and integrated interpretation across the complete severity continuum.

Integrated Approaches to Testing and Assessment (IATA) require transparent, method-independent reference datasets that integrate regulatory classifications, in vivo metrics, and in vitro outcomes within a unified severity-based framework. The absence of such harmonized datasets limits the systematic evaluation of method performance, impedes the identification of borderline substances, and restricts severity-stratified analysis. The integration of in vitro RhE models within an IATA framework represents a shift toward

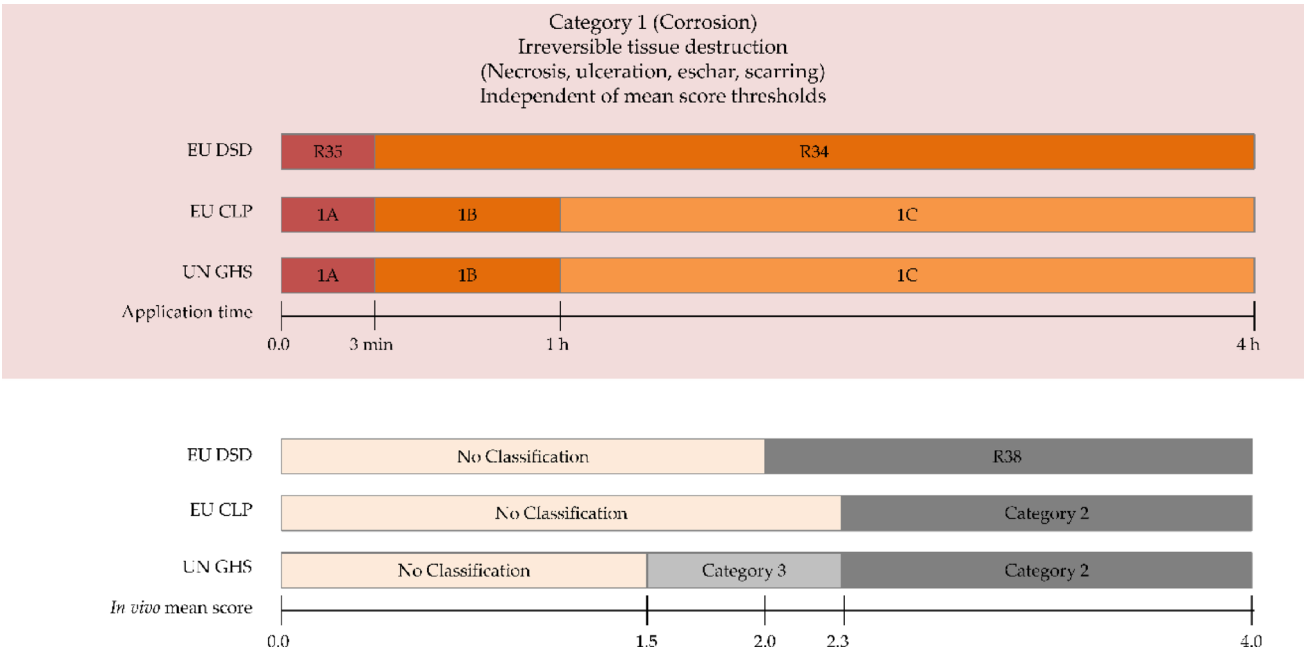


Fig. 1 Regulatory comparison of skin corrosion and irritation classification criteria under EU DSD, EU CLP, and UN GHS. The figure presents the severity continuum from non-irritating to corrosive substances. Category 1 subcategories (**A**–**C**) are defined based on the onset of irreversible effects, whereas Categories 2 and 3 are determined by mean erythema and edema scores. This framework provides the basis for integrating in vivo and in vitro data within the ChemSkin 2.0 database

mechanistic toxicology, leveraging NAMs to provide human-relevant safety data. The utilization of such mechanistic pathways (e.g., Adverse Outcome Pathways, AOPs) allows for a more robust prediction of toxicological effects compared to traditional apical endpoints observed in animal studies (Leist et al. 2017).

We established the ChemSkin database (Han et al. 2021) primarily focused on curated skin irritation data to support the comparative evaluation of RhE models for the test of skin irritation. Although limited corrosion information was included in ChemSkin DB, a comprehensive representation of corrosion severity was not a central objective of that initial version. This limitation underscored the critical necessity for an expanded reference resource capable of systematically integrating both corrosion and irritation endpoints within a unified regulatory framework.

To address this gap, ChemSkin 2.0 was developed as a harmonized reference chemical database integrating skin irritation and corrosion data across the complete regulatory severity spectrum. The database expands chemical coverage, strengthens the representation of Category 1 corrosives with explicit 1A/1B/1C stratification, and aligns classification assignments according to UN GHS criteria while preserving traceability to EU CLP and historical systems. Structured fields include chemical identity, regulatory classifications, normalized in vivo Draize metrics, human evidence, and corresponding in vitro outcomes derived from validated OECD test guidelines.

By encompassing the entire corrosion–irritation continuum, ChemSkin 2.0 provides a transparent and severity-oriented reference resource intended to support the regulatory interpretation of in vitro data and facilitate IATA development. The present study describes the construction, harmonization strategy, and structural characteristics of the database and evaluates its applicability as a comprehensive reference framework for skin hazard assessment.

Materials and methods

Data sources and chemical inclusion criteria

Reference chemicals were identified through systematic searches of peer-reviewed literature and publicly available regulatory resources published up to December 2025. Bibliographic searches were conducted in PubMed, Scopus, and Web of Science using combinations of keywords including “skin irritation”, “skin corrosion”, “OECD TG 404”, “OECD TG 431”, “OECD TG 439”, “reconstructed human epidermis”, and “validation study”.

Studies reporting skin irritation or corrosion data were screened, yielding 2,680 initial data entries. After consolidation of chemical names and removal of duplicates based on CAS Registry Numbers, 668 candidate substances were identified.

Candidate substances were evaluated according to predefined eligibility criteria. Chemicals were included when: (i) a unique CAS Registry Number corresponding to a defined substance was available, and (ii) sufficient information existed to assign a skin corrosion or irritation category. This information consisted of either in vivo skin irritation or corrosion data reported with sufficient experimental detail, or a harmonized classification under the EU CLP Regulation.

Substances were excluded when in vivo data were absent, insufficiently described or ambiguous and no harmonized classification was available. Following application of these criteria, 234 final substances were included in the ChemSkin 2.0 database.

Regulatory classification harmonization

Regulatory classifications were harmonized to enable consistent severity interpretation across regulatory systems. When a harmonized classification for skin corrosion or irritation was available under Annex VI of Regulation (EC) No 1272/2008 (European Union 2008b), this classification was used as the primary regulatory reference.

For severity-stratified analysis, final reference categories were aligned with the United Nations Globally Harmonized System (UN GHS), which distinguishes No Category, Category 3 (mild irritation), Category 2 (irritation), and Category 1 subcategories (1A, 1B, 1C). While CLP harmonized classifications served as the primary regulatory reference, GHS terminology was used for severity stratification.

Governmental GHS classifications were identified through searches of the OECD eChemPortal. Through this platform, national classifications, including those from the Japanese National Institute of Technology and Evaluation (NITE) were retrieved when available. When multiple sources were available, classification priority followed a hierarchical approach: (i) harmonized CLP classification, (ii) evaluation of in vivo experimental data, and (iii) governmental GHS classifications. The hierarchical workflow used to assign final reference classifications is summarized in Fig. 2.

In vivo data extraction and severity-based classification

In vivo rabbit skin irritation and corrosion data were primarily derived from studies conducted according to OECD Test Guideline 404 or equivalent protocols, including EPA OPPTS 870.2500 (US EPA 1998) and EU Method B.4

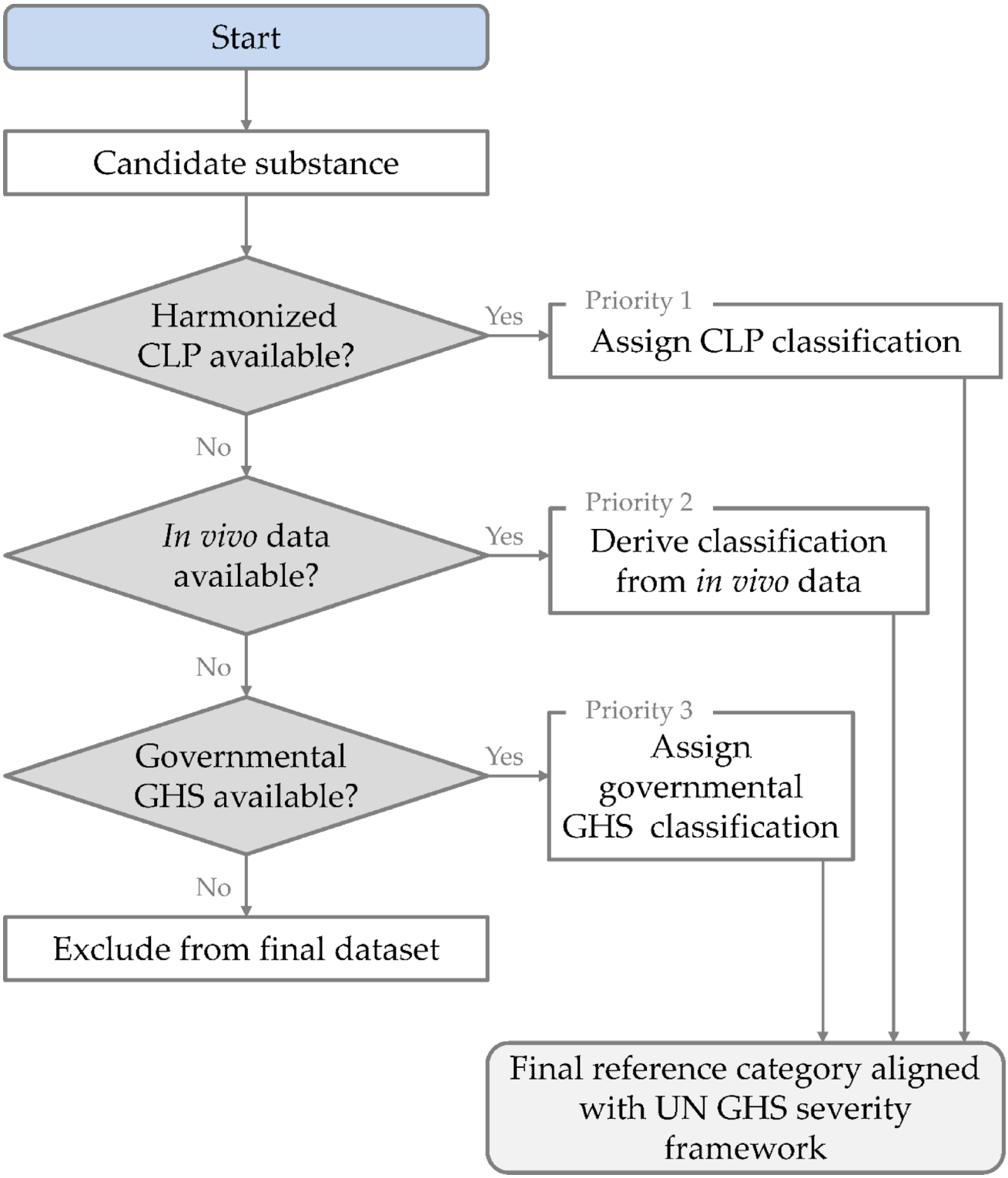


Fig. 2 Workflow for assigning final reference categories in the ChemSkin 2.0 database. Harmonized classifications from Annex VI of the EU CLP Regulation were used as the primary reference (Priority 1). In the absence of harmonized classifications, categories were assigned based on in vivo experimental data (Priority 2) or governmental GHS classifications retrieved via OECD eChemPortal (Priority 3). This hierarchical approach ensures consistent and traceable classification across the dataset

(European Union 2008a), provided that study design and scoring criteria were comparable.

For each study, erythema and edema scores at 24, 48, and 72 h were extracted, and the dominant irritation score was defined as the higher of the mean erythema or mean edema values across the 24–72 h observation window. Primary Dermal Irritation Index (PII) values, when available, were recorded as supplementary information.

Substances were classified as corrosive when irreversible tissue destruction (e.g., necrosis, ulceration, eschar formation, or full-thickness dermal damage) was reported, independent of irritation scores. For reversible responses, severity categories were assigned according to the GHS-based classification framework described in the Introduction.

In vitro data integration (OECD TG 431 and 439)

Reconstructed human epidermis (RhE)-based in vitro data were collated from studies conducted according to OECD TG 431 (skin corrosion) and OECD TG 439 (skin irritation). For both test guidelines, cell viability values obtained under guideline-recommended exposure conditions were systematically extracted. When multiple laboratories or studies reported results for the same substance–model combination, minimum and maximum viability values were recorded to capture inter-laboratory variability and experimental dispersion.

To maintain a conservative interpretation within the Integrated Approach to Testing and Assessment (IATA) framework, model-specific predicted classifications were derived using the minimum reported viability value for each substance–model combination. This decision rule was applied consistently across both corrosion and irritation evaluations to minimize the risk of hazard underestimation arising from experimental variability. Such a protective interpretation strategy is consistent with the data integration principles described in the OECD Guidance Document No. 203 (OECD 2014) and the regulatory data evaluation framework outlined in the ECHA guidance on information requirements (ECHA 2017).

For OECD TG 431, corrosion classification and, where applicable, subcategorization (Category 1A versus 1B/1C) were determined according to guideline-defined exposure durations and viability thresholds. The minimum viability value corresponding to each relevant exposure time served as the primary decision reference for category assignment.

For OECD TG 439, predicted irritation classification (Irritant vs. Non-irritant) was determined relative to the guideline-defined 50% viability threshold, using the minimum reported viability value as the decision anchor.

Database structure

The ChemSkin 2.0 database was organized into three interlinked components designed to support transparent reference categorization and the systematic management of model-specific experimental data.

The first component, the Substance Reference Table, serves as the central backbone of the database. This table contains chemical identity (name and CAS Registry Number), physical state, chemical category, and the reference category assigned according to the hierarchical regulatory framework described in Fig. 2. Relevant in vivo data and available human data are recorded in the database. Each substance was assigned a unique internal identification number (ID), which is consistently applied across all components to enable direct cross-referencing between reference categories and in vitro records.

The second component comprises the OECD TG 431 corrosion records. For each substance ID and RhE model, exposure duration and cell viability values are documented as model-specific minimum–maximum ranges when multiple experimental results were available. This structure preserves inter-laboratory variability within each model. Predicted corrosion classifications used for performance evaluation were derived from the minimum viability value according to guideline-defined thresholds.

The third component consists of the OECD TG 439 irritation records. For each substance ID and RhE model, cell viability values are likewise recorded as model-specific minimum–maximum ranges. Predicted irritation classifications were determined using the minimum reported viability value relative to the 50% threshold defined in OECD TG 439. These predicted classifications are maintained as separate analytical layers rather than being displayed within the primary database tables.

Statistical and descriptive analyses

Descriptive statistics were utilized to summarize the distribution of substances across reference categories and predefined chemical groups. The proportional representation of Category 1 subcategories (1A, 1B, and 1C), Category 2, Category 3, and Not Classified (NC) was calculated to characterize database coverage across the complete corrosion–irritation spectrum.

The binary predictive performance of TG 431 and TG 439 models relative to the reference categories was evaluated using standard performance metrics, including sensitivity, specificity, and overall accuracy. Cohen's kappa coefficient (κ) was calculated to assess the level of agreement between in vitro predictions and reference data beyond that expected

by chance. These metrics were derived independently for each Reconstructed Human Epidermis (RhE) model.

For substances with explicit Category 1 subcategorization, we quantified exact subcategory agreement, the frequency of subcategory shifts, and the incidence of underprediction (i.e., corrosive substances misclassified as non-corrosive). To evaluate the preservation of the ordinal severity structure, Spearman's rank correlation coefficients (ρ) were calculated between the reference severity rankings and model-derived outcomes. In the TG 439 irritation analysis, the relationship between in vivo irritation severity and RhE cell viability was similarly examined using Spearman's rank correlation to assess the quantitative-to-qualitative correspondence.

To identify category-dependent variability in model performance, predictive accuracy was stratified by predefined chemical groups. Associations between chemical categories and prediction outcomes (e.g., correct vs. incorrect) were evaluated using Pearson's chi-square test ($\chi 2$).

All statistical analyses were conducted using IBM SPSS Statistics (Version 23.0, IBM Corp., Armonk, NY, USA). Statistical significance was defined as $p < 0.05$. No multivariable inferential modeling was applied, consistent with the objective of providing a transparent and empirical reference framework for evaluating in vitro model performance.

Results and discussion

Database composition and severity distribution

The ChemSkin 2.0 database includes 234 substances covering the full spectrum of skin corrosion and irritation. Each substance was assigned a unique internal identification number (ID) based on its CAS Registry Number to ensure consistent cross-referencing between OECD TG 431 and TG 439 in vitro outcomes.

Substances were grouped into seven chemical categories representing structural and mechanistic diversity. Reference categories were established by integrating regulatory classifications (EU CLP and GHS), in vivo Draize metrics, and available human data.

A comprehensive overview of substance information is provided in the Supplementary Electronic Material (Table S1).

Severity distribution across reference category

The distribution of reference categories across the corrosion–irritation spectrum is presented in Fig. 3. Not Classified (NC) substances account for 46% of the database. Category 2 (Irritant) and Category 3 (Mild Irritant) constitute 22% and 11%, respectively. Corrosive substances (Category 1 and its subcategories) comprise 21% of the database.

Within the corrosive group, the database includes Category 1A (6%), 1B (6%), 1C (2%), and Category 1 without subclass differentiation (7%). This distribution includes varying degrees of irreversible tissue damage to enable severity-resolved evaluation.

Physical state composition and chemical categories

The distribution of physical states is shown in Fig. 4a. Liquids comprise 68% of the database, followed by solids (31%) and gels (1%). This composition reflects the predominance of liquid substances in regulatory datasets while

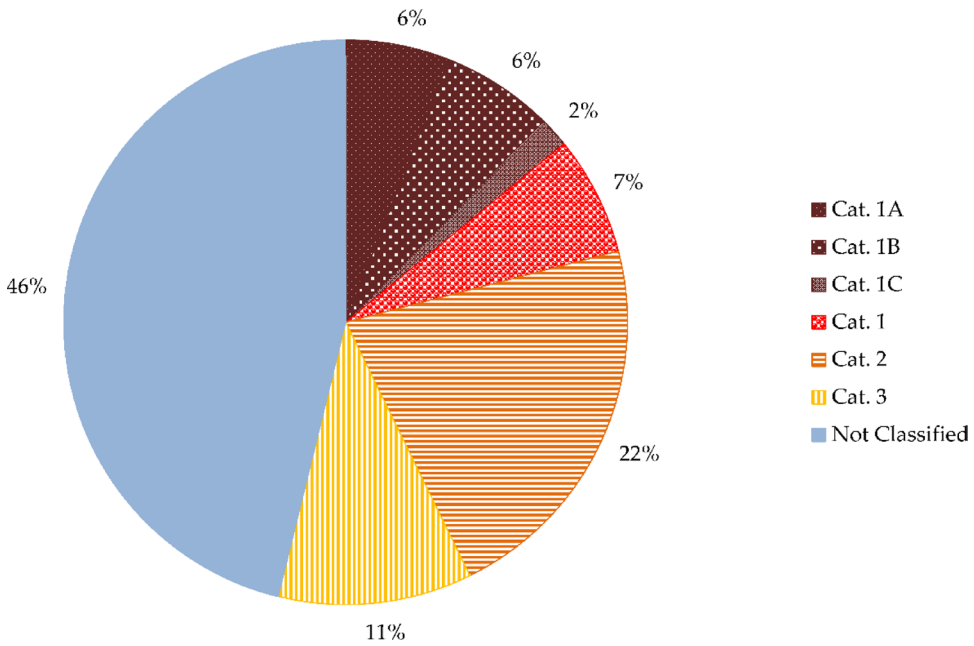


Fig. 3 Distribution of reference skin corrosion and irritation classifications across the severity spectrum. The bar chart shows the chemical coverage of the ChemSkin 2.0 database according to UN GHS and EU CLP classifications, including No Category (NC), Category 2/3 (irritants), and Category 1 subcategories (1A, 1B, and 1C; corrosives). The distribution highlights the expanded representation of corrosive substances and detailed stratification within Category 1

maintaining representation of solid materials for evaluating the applicability domain of RhE models.

Figure 4b presents the distribution of substances across predefined chemical categories. Reactive organics constitute 39% of the database, followed by neutral organic solvents (31%). Strong acids and bases account for 14%, while inorganic compounds (6%), aromatic sulfonates/dyes/salts (5%), surfactants (4%), and others (1%) comprise the remainder.

This distribution indicates that the database includes diverse structural and chemical categories to support the evaluation of category-specific performance patterns.

The RhE-based skin corrosion test (OECD TG 431) database

In vitro skin corrosion data generated according to OECD TG 431 were compiled for five validated reconstructed human epidermis (RhE) models: EpiSkin™ (SM), EpiDerm™ SCT (EPI-200), SkinEthic™ RHE, epiCS®, and LabCyte EPI-MODEL24 SCT.

The Supplementary Electronic Material (Table S2) summarizes the TG 431 records for each substance ID, including model-specific exposure durations and corresponding cell viability values. In cases where multiple independent studies were available for a given substance, all reported outcomes were retained and presented as minimum–maximum viability ranges.

This representation captures inter-study variability while maintaining traceability to the underlying experimental data. Derived classifications were not embedded in the database tables, allowing a clear separation between raw measurements and subsequent performance evaluation.

Performance of TG 431 models relative to the reference category

Figures 5 and 6 present the distribution of RhE cell viability values across reference corrosion categories for the five OECD TG 431 models.

Corrosive substances (Category 1A–1C) consistently exhibited low viability, while non-corrosive substances (Category 2, Category 3, and NC) showed higher viability distributions across all models. For performance evaluation, predicted corrosion classifications were derived from the minimum reported viability values according to guideline-defined exposure times and thresholds.

All five RhE models demonstrated high concordance with the reference categories (Table 1). Overall accuracy ranged from 81.2 to 86.1%, and Cohen's kappa coefficients ranged from 0.622 to 0.720, indicating substantial agreement. The elevated kappa values suggest that the agreement reflects meaningful alignment with the severity-based reference structure rather than being driven by class prevalence.

Sensitivity for identifying corrosive substances (Category 1) was high across all models, ranging from 97.3 to 100.0%. EpiSkin™ (SM), EpiDerm™ SCT, and LabCyte EPI-MODEL24 SCT achieved 100% sensitivity. This high sensitivity indicates the robustness of RhE systems in detecting irreversible tissue damage.

The distribution of correct and incorrect classifications for corrosive and non-corrosive substances is shown in Fig. 7. Specificity ranged from 61.9 to 77.6%, with SkinEthic™ RHE achieving the highest and EpiDerm™ SCT the lowest (Fig. 7b). The lower specificity observed in some models is associated with borderline substances—primarily Category 2 and Category 3 entries—located near the corrosion decision threshold within the regulatory severity continuum.

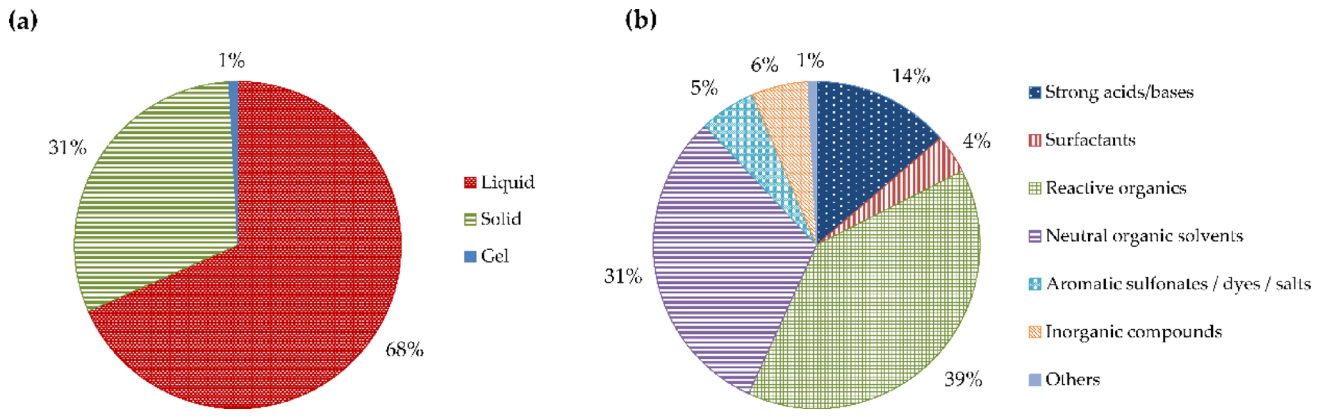


Fig. 4 Distribution of substances in the ChemSkin 2.0 database according to **a** physical states and **b** structural-mechanistic chemical categories. The dataset includes a diverse range of physical states and chemical categories, supporting comprehensive evaluation across different substance types

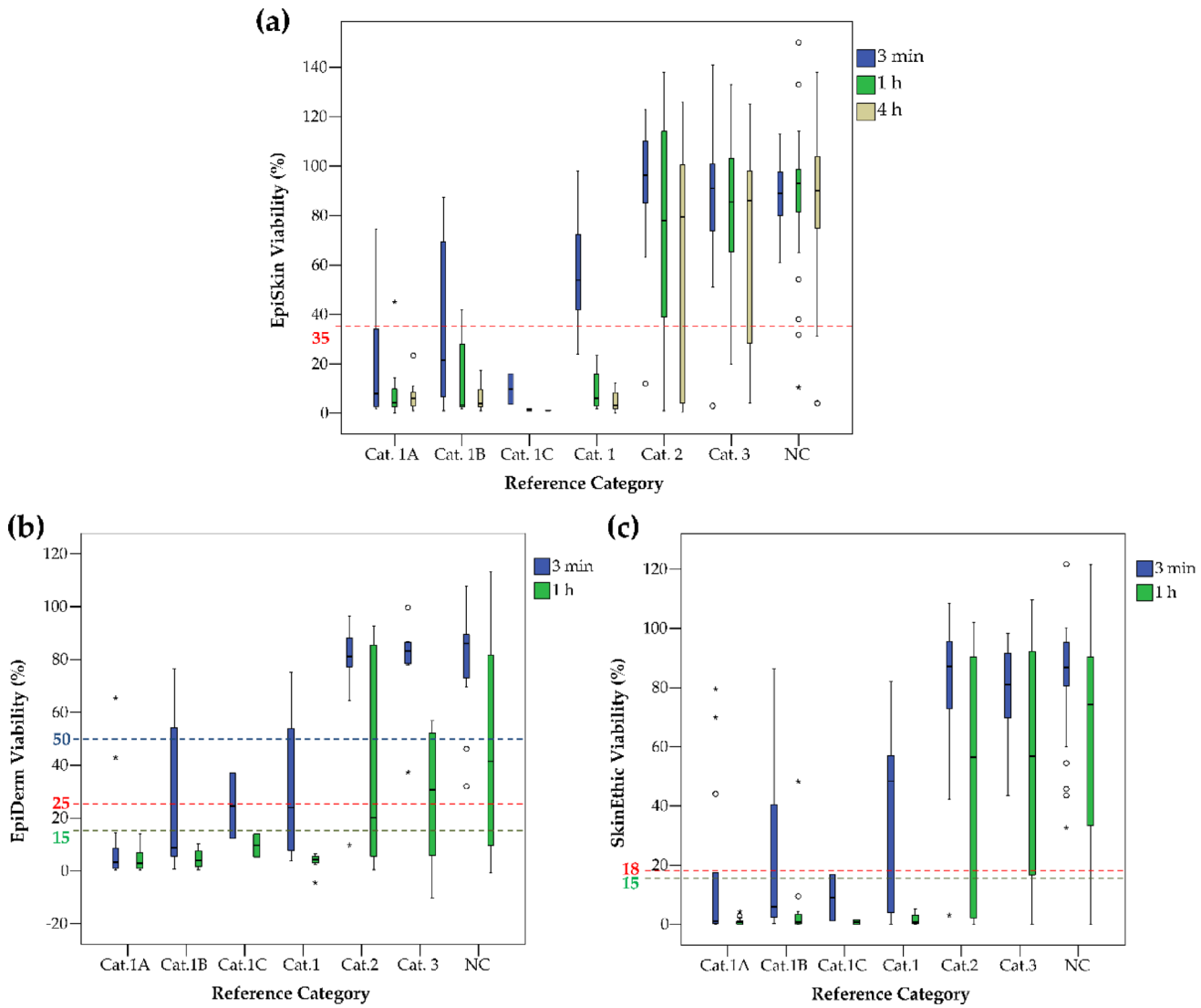


Fig. 5 Distribution of RhE cell viability values across reference corrosion categories for three OECD TG 431 skin corrosion models. **a** EpiSkin™ (SM); **b** EpiDerm™ SCT (EPI-200); **c** SkinEthic™ RHE. Box plots represent the distribution of cell viability values for each reference category (Cat. 1A, 1B, 1C, Cat. 1, Cat. 2, Cat. 3, and NC). Horizontal dashed lines indicate the decision thresholds used for corrosion classification

Overall, TG 431 models demonstrated reliability for identifying corrosive substances. The variability in specificity is primarily linked to substances positioned near regulatory classification boundaries rather than intrinsic model limitations. This pattern is consistent with the conservative hazard identification framework of OECD TG 431, which prioritizes the detection of corrosive hazards, potentially leading to lower specificity when borderline irritants occur near the decision threshold.

Subcategory-level performance (category 1A–1C)

To evaluate whether RhE models preserve the hierarchical potency structure within Category 1, a severity-resolved analysis was conducted for substances with defined subcategorization (1A–1C) (Table 2).

Exact subcategory accuracy ranged from 53.8 to 60.7%, indicating that approximately half of the corrosive substances were assigned to the correct reference subcategory. A substantial proportion of predictions involved subcategory shifts (39.1–42.3%), where substances were correctly identified as corrosive but assigned to a different Category 1 subcategory. Notably, underprediction to non-corrosive outcomes was rare (0.0–4.3%), demonstrating that misclassifications occurred predominantly within the corrosive domain rather than as false negatives.

Spearman rank correlation analysis showed positive monotonic associations between reference severity rankings

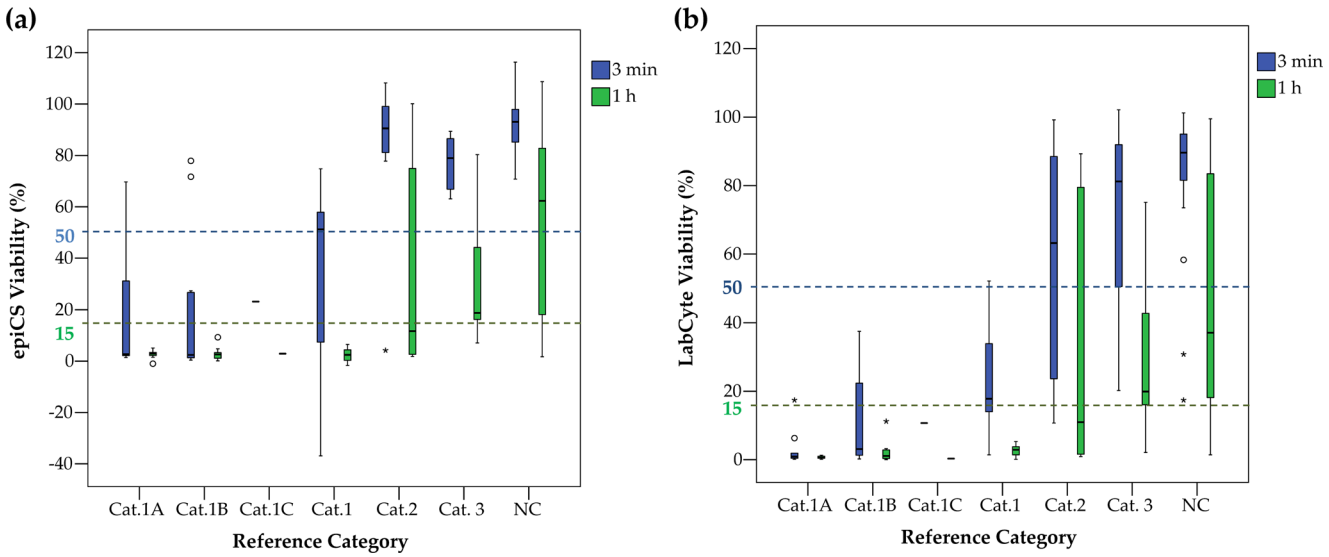


Fig. 6 Distribution of RhE cell viability values across reference corrosion categories for two additional OECD TG 431 skin corrosion models. **a** epiCS®; **b** LabCyte EPI-MODEL24 SCT. The data representation and decision thresholds are consistent with those described in Fig. 5

Table 1 Predictive performance metrics of OECD TG 431 RhE models relative to reference categories

RhE Model	N	Sensitivity (%)	Specificity (%)	Accuracy (%)	Kappa
EpiSkin™ (SM)	106	100.0	74.2	84.0	0.685
EpiDerm™ SCT (EPI-200)	85	100.0	61.9	81.2	0.622
SkinEthic™ RHE	116	97.5	77.6	84.5	0.686
epiCS®	72	97.3	74.3	86.1	0.720
LabCyte EPI-MODEL24 SCT	73	100.0	68.4	83.6	0.675

N number of substances evaluated per model, Sensitivity, proportion of reference corrosive substances correctly predicted; Specificity, proportion of reference non-corrosive substances correctly predicted; Accuracy, overall proportion of correct predictions; Kappa, Cohen's kappa coefficient. Performance indicators were calculated against the reference category using model-specific predicted classifications derived from minimum viability values

(1C < 1B < 1A) and model prediction scores ($\rho = 0.201$–0.593). The strongest ordinal concordance was observed for LabCyte EPI-MODEL24 SCT ($\rho = 0.593$, $p = 0.004$), whereas epiCS® showed a weaker and non-significant association ($\rho = 0.201$, $p = 0.358$). These findings indicate partial preservation of the corrosion severity gradient across most models.

Collectively, these results indicate that TG 431 RhE systems are optimized for corrosion hazard identification, while discrimination among Category 1 subcategories shows moderate ordinal concordance that varies by model. This pattern is consistent with the exposure-time–dependent threshold structure of OECD TG 431, which supports categorical decision-making but may compress potency distinctions near boundary values. The findings highlight that while TG 431 models are effective for hazard identification, the fine resolution of corrosive potency within Category 1 subcategories remains limited.

The RhE-based skin irritation test (OECD TG 439) database

In vitro skin irritation data generated according to OECD TG 439 were compiled for the ChemSkin 2.0 database using seven validated reconstructed human epidermis (RhE) models: EpiSkin™, EpiDerm™, SkinEthic™, LabCyte EPI-MODEL24, epiCS®, Skin + ®, and KeraSkin™.

Table 3 summarizes the TG 439 records for each substance ID, including model-specific cell viability values expressed as minimum–maximum ranges. For substances with multiple independent studies, all reported experimental outcomes were retained to preserve the full observed range. This representation captures inter-study variability while maintaining traceability to the underlying experimental data. Predicted irritation classifications were derived separately based on the guideline-defined 50% viability threshold.

Performance of TG 439 models relative to the reference category

Figures 8 and 9 present the relationship between in vivo irritation scores and RhE cell viability values across the seven OECD TG 439 models. Substances classified as irritants (Category 2) generally showed viability values below the 50% threshold, while non-irritant substances—including Category 3 substances consolidated as non-classified

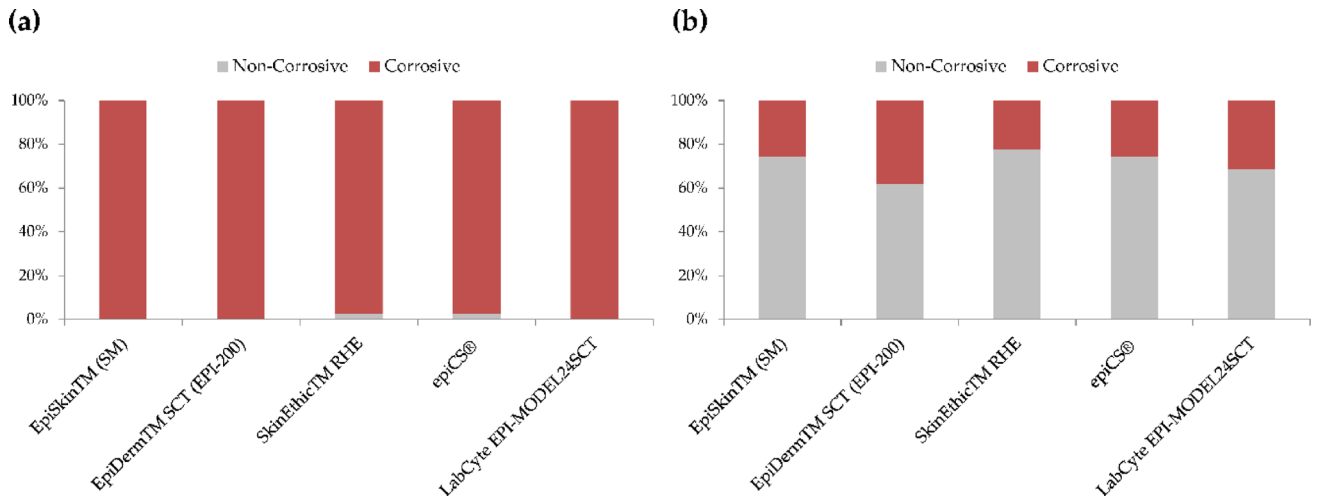


Fig. 7 Predictive classifications of OECD TG 431 RhE models relative to corrosive and non-corrosive substances. **a** Predicted outcomes for substances classified as corrosive according to the reference category; **b** predicted outcomes for substances classified as non-corrosive according to the reference category

Table 2 Severity-resolved performance of OECD TG 431 RhE models for Category 1 subcategorization (1A-1C)

RhE Model	N (1A-1C)	Accuracy (%)	Subcategory Shift (%)	Underprediction (%)	Spearman's ρ	*p* -value
EpiSkin™ (SM)	26	57.7	42.3	0.0	0.435	0.026*
EpiDerm™ SCT (EPI-200)	28	60.7	39.3	0.0	0.477	0.010*
SkinEthic™ RHE	26	53.8	42.3	3.8	0.497	0.010*
epiCS®	23	56.5	39.1	4.3	0.201	0.358
LabCyte EPI-MODEL24 SCT	24	59.1	40.9	0.0	0.593	0.004**

N (1A–1C), number of substances with defined Category 1 subcategorization (1A–1C) included for each RhE model; Accuracy (%), proportion of substances for which the predicted classification matched the exact reference subcategory (1A, 1B, or 1C); Subcategory shift (%), proportion predicted as corrosive but assigned to a different Category 1 subcategory relative to reference; Underprediction (%), proportion of reference Category 1 substances predicted as non-corrosive. Spearman's ρ, rank correlation coefficient between reference subcategory severity and model prediction score; p-value, two-tailed significance level (*$p<0.05$, **$p<0.01$.). Percentages may not sum to 100 due to rounding

(NC) under EU CLP—tended toward higher viability distributions. However, substances with in vivo scores near regulatory classification boundaries were distributed around the 50% viability threshold. To quantify this relationship, Spearman rank correlation analysis revealed significant negative correlations across all evaluated models (Table 4).

The results indicate that increasing in vivo irritation severity is consistently associated with reduced RhE cell viability. While the strength of correlation varied ($\rho = -0.523$ to -0.778), all relationships remained statistically significant at the 0.01 level. This quantitative alignment supports the biological relevance of RhE viability as an indicator of irritation severity.

Predictive performance metrics relative to the reference categories are summarized in Table 5, with outcome distributions presented in Fig. 10.

Overall accuracy ranged from 66.4 to 100.0%, and Cohen's kappa coefficients varied between 0.339 and 1.000, indicating moderate to substantial agreement. Compared with the corrosion analysis (TG 431), kappa values for TG 439 were generally lower and exhibited greater variability, reflecting the increased uncertainty inherent to irritation prediction.

Sensitivity for identifying Category 2 irritants ranged from 80.6 to 100.0%. As shown in Fig. 10a, the majority of reference irritants were correctly predicted, supporting the reliability of TG 439 systems for hazard detection. In contrast, specificity for non-irritant substances showed greater variability, ranging from 60.2 to 100.0% (Fig. 10b). This tendency toward overprediction in several models contributes to the lower kappa values compared to TG 431.

This variability is consistent with the structural characteristics of TG 439. Unlike TG 431, which uses exposure-time–dependent thresholds, TG 439 relies on a single 50% cell viability cutoff for binary classification. Under EU CLP, the consolidation of Category 3 (mild irritation) into the NC group effectively compresses the irritation severity gradient. Consequently, substances with viability values near the 50% threshold are susceptible to classification shifts due to modest experimental variation.

The preservation of minimum–maximum viability ranges within the database enables the examination of these threshold-adjacent cases. By maintaining separation

Table 3 OECD TG 439 (Skin Irritation) RhE model viability records for substances in the ChemSkin 2.0 database

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
1	Formic acid	64–18-6	Cat. 1A	4.3–4.6		0.7–1.2					Alepee et al. (2015b, 2019)
2	Acetic acid	64–19-7	Cat. 1A	3.5–5.8		0.6–1					Alepee et al. (2015b, 2019)
3	1,2-Diaminopropane	78–90-0	Cat. 1A	1.8–10.8		0.3–0.6					Alepee et al. (2015b, 2019)
4	Bromoacetic acid	79–08-3	Cat. 1A	3.3–5.0		0.4–1.1					Alepee et al. (2015b, 2019)
5	Acrylic acid	79–10-7	Cat. 1A	3.1–5.6		0.3–0.6					Alepee et al. (2015b, 2019)
6	Dichloroacetyl chloride	79–36-7	Cat. 1A	7.7–10.6		0.6–0.9					Alepee et al. (2015b, 2019)
7	Potassium hydroxide (5% aq.)	1310–58-3	Cat. 1A	11.7–47.6	1.4–11.5	0.1–2.6	2.5–3.4	0.6–14.5	5.2	-4.48–8.1	European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2006), Katoh and Hata (2011), OECD (2025a, b, c)
9	8 N KOH	1310–58-3	Cat. 1A	6.8–22.8		0.2–0.4					Alepee et al. (2015b, 2019)
11	Hydrochloric acid (14.4%)	7647–01-0	Cat. 1A	41.1–50.6		0.4–0.8					Alepee et al. (2015b, 2019)
12	Sulphuric acid (98%)	7664–93-9	Cat. 1A	6.8–10.0		0.7–1					Alepee et al. (2015b, 2019)
13	Hydrobromic acid (48%)	10,035–10-6	Cat. 1A	5.1–10.6		0.9–1					Alepee et al. (2015b, 2019)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
14	Dimhyldipropylene-triamine (N,N-Dimhyldipropylenriamine)	10,563–29-8	Cat. 1A	7.3–23.2		0.5–0.9					Alepee et al. (2015b, 2019)
15	Propionic acid	79–09-4	Cat. 1B	3.0–5.5		0.6–0.7					Alepee et al. (2015b, 2019)
16	N,N-Dimethylbenzylamine	103–83-3	Cat. 1B	4.7–6.8		0.5–0.9					Alepee et al. (2015b, 2019)
17	Butyric acid	107–92-6	Cat. 1B	2.1–4.5		0.4–1.1				2.2	Alepee et al. (2015b, 2019)
18	Maleic anhydride	108–31-6	Cat. 1B	4.7–7.5		0.4–0.5					Alepee et al. (2015b, 2019)
19	Cyclohexylamine	108–91-8	Cat. 1B	4.3–6.6		0.5–1.9					Alepee et al. (2015b, 2019)
20	Phenol	108–95-2	Cat. 1B	4.1–6.1		0.5–0.6					Alepee et al. (2015b, 2019)
21	Heptanoic acid	111–14-8	Cat. 1B							2.2	Han et al. (2021)
23	Ethanolamine	141–43-5	Cat. 1B			0.2–21.2					Alepee et al. (2015b)
24	Methylisothiazolinone	2682–20-4	Cat. 1B	2.4–3.1							Alepee et al. (2016)
25	Sodium metasilicate(10%)	6834–92-0	Cat. 1B	85.60							Cotovio et al. (2005)
27	Phosphorus tribromide	7789–60-8	Cat. 1B	5.8–12.6		0.4–1.4					Alepee et al. (2015b, 2019)
29	Fluoboric acid (hydrogentetrafluoroborate) (48%)	16,872–11-0	Cat. 1B	4.7–5.1		0.6–1.1					Alepee et al. (2015b, 2019)
30	25% Cetyltrimethylammonium chloride solution (in D.W.)	112–02-7	Cat. 1C	3.7–23.5						1.1	Alepee et al. (2015b, 2019)
31	Steartrimonium Chloride	112–03-8	Cat. 1C	19.7	7.4	12.1	24.9				Sugiyama et al. (2018)
32	Octanoic acid (caprylic acid)	124–07-2	Cat. 1C	4.5–6.1		0.5–1.1				2.0	Alepee et al. (2015b, 2019)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
34	Lactic acid	50–21-5	Cat. 1			0.5–1.5					Alepee et al. (2015b)
35	Methacrolein	78–85-3	Cat. 1	6.2–7.2		0.9–1.5					Alepee et al. (2015b, 2019)
36	2-tert-Butylphenol	88–18-6	Cat. 1	3.1–5.3		1.4–9.5					Alepee et al. (2015b, 2019)
37	Allyl bromide	106–95-6	Cat. 1	6.4–9		0.7–1.7					Alepee et al. (2015b, 2019)
38	n-Heptylamine	111–68-2	Cat. 1A	2.8–5.5		0.6–4.4					Alepee et al. (2015b, 2019)
39	Boron trifluoride acetic acid complex	373–61-5	Cat. 1	3.4–4.5		0.4–1.0					Alepee et al. (2015b, 2019)
40	Carvacrol	499–75-2	Cat. 1	4.5–5.6		0–27.1					Alepee et al. (2015b, 2019)
41	N,N-Dimethylisopropylamine	996–35-0	Cat. 1	6.3–8.4		0.4–1.3					Alepee et al. (2015b, 2019)
42	Glycol bromoacetate (85%)	3785–34-0	Cat. 1	3.8–5.8		0.4–1					Alepee et al. (2015b, 2019)
43	3-Methoxypropylamine	5332–73-0	Cat. 1	3.9–10.8		0.3–0.5					Alepee et al. (2015b, 2019)
44	Phosphoric acid	7664–38-2	Cat. 1	26.6–54.1		0.5–1.2					Alepee et al. (2015b, 2019)
46	Origanum oil	8007–11-2	Cat. 1			0.89					Fiume et al. (2016)
47	60/40 octanoic/decanoic acid(Caprylic/Capric acids)	68,937–75-7	Cat. 1	3.2–5.7		1–1.3					Alepee et al. (2015b, 2019)
48	55/45 octanoic/decanoic acid(Caprylic/Capric acids)	68,937–75-7	Cat. 1	4.5–7.3		0.7–1.1					Alepee et al. (2015b, 2019)
49	65/35 octanoic/decanoic acid(Caprylic/Capric acids)	68,937–75-7	Cat. 1	4.7–6.3		0.7–1.3					Alepee et al. (2015b, 2019)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
50	Benzoic acid	65–85-0	Cat. 2	101–104.3							Alepee et al. (2016)
51	1,1,1-Trichloroethane	71–55-6	Cat. 2	36.61–90.0	6.6–23.52	2.9–12.5	10.1–13.4				Kandárová et al. (2009), Kandárová et al. (2006), OECD (2011a), Zuang et al. (2002)
52	Terpinyl acetate	80–26-2	Cat. 2	4.9–77.9	10.16–81.53	1.36–5.05	18.0–99.5				Alepee et al. (2015b, 2019)
53	Lilestralis/Lilial	80–54-6	Cat. 2	11.3–55.56	7.4–12.8	1.3–2.6					Cotovio et al. (2005), Kandárová et al. (2004), Kandárová et al. 2006; Zuang et al. 2002)
54	2-Methoxyphenol (guaiacol)	90–05-1	Cat. 2			0.5–0.6					Alepee et al. (2015b)
55	Butyl methacrylate	97–88-1	Cat. 2	6.8–35.6	8.5–98.11	0.95–4.31	22.5–66.3		24.6		Alepee et al. (2019), Alépée et al. (2010), Han et al. (2021), Kandárová et al. (2009), OECD (2011b), Spielmann et al. (2007)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
56	alpha-Terpineol	98–55-5	Cat. 2	2.75–15.16	6.0–70.87	0.65–7.13	4.97–33.0		17.1		Alépée et al. (2010), Han et al. (2021), Kandárová et al. (2009), Katoh et al. (2009), OECD (2011a), Spielmann et al. (2007)
57	Cyclamen aldehyde	103–95-7	Cat. 2	6.9–51.1	8.67–26.39	1.59–1.83	7.2–36.0	1.2–82.1	3.0	-5.85–15.6	Alepee et al. (2015b, 2019), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Katoh and Hata (2011), OECD (2011b), OECD (2025a, b, c), Spielmann et al. (2007)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
58	Cinnamaldehyde	104–55-2	Cat. 2	4.2–7.49	4.09–12.4	1.2–1.7	21.4–35.6	0.5–3.6		-1.93–20.0	Alepee et al. (2015b, 2019), Cotovio et al. (2005), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Kandárová et al. (2005), Katoh and Hata (2011)
59	1-Bromopentane	110–53-2	Cat. 2	17.0–91.0	4.43–89.2	1.3–2.9	17.7–24.3			11.6	Han et al. (2021), Kandárová et al. (2009, 2004, 2006), OECD (2011a), Zuang et al. (2002)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
60	1-Bromohexane	111–25-1	Cat. 2	9.5–53.6	12.7–100.41	0.7–3.1	16.9–89.1	5.3–105.7	5.6	4.77–56.8	Alepee et al. (2019), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Kandárová et al. (2006), Katoh et al. (2009), OECD (2011b), OECD (2025a, b, c), Spielmann et al. (2007)
61	Heptanal	111–71-7	Cat. 2	8.2–29.44	4.72–14.2	1.07–1.59	8.0–23.3	0.6–1.6	9.3	-1.29–17.4	Alépée et al. (2010), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Kandárová et al. (2004), OECD (2011b), OECD (2025a, b, c), Zuang et al. (2002)
62	Methyl laurate	111–82-0	Cat. 2	105.38	100.2–107.4					91.2	Cotovio et al. (2005), Han et al. (2021), Kandárová et al. (2005)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
63	Nonanoic acid	112–05-0	Cat. 2			1.06				3.0	Han et al. (2021), Tornier et al. (2010)
64	1-Decanol	112–30-1	Cat. 2	5.2–8.0	4.47–58.65	1.24–2.4	8.2–42.1	1.3–100		2.28–16.1	Alepee et al. (2019), Alepee et al. (2015b), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Katoh and Hata (2011), OECD (2011b), Spielmann et al. (2007))
65	Methyl palmitate	112–39-0	Cat. 2	69.5–106.32	81.0–117.0	70.9–122.7					Alepee et al. (2015b), Cotovio et al. (2005), Kandárová et al. (2004, 2006), Zuang et al. (2002)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
66	Tetrachloroethylene	127–18-4	Cat. 1	7.5–26.20	4.2–15.7	0.6–1.5	11.1–22.6	0.8–11.3		2.94–51.7	Alepee et al. (2019), Alepee et al. (2015b), Cotovio et al. (2005), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Kandárová et al. (2005), Katoh and Hata (2011)
67	Diethyl toluamide	134–62-3	Cat. 2	85.5							Alepee et al. (2015a)
68	Sodium lauryl sulphate (50% aq.)	151–21-3	Cat. 1	4.8–77.1	2.0–14.5	0.6–2.8					Kandárová et al. (2004, 2006), Zuang et al. (2002)
69	Sodium lauryl sulphate (20% aq.)	151–21-3	Cat. 1	5.2–61.32	3.38–21.2	1–1.7					Alepee et al. (2019, 2015b), Cotovio et al. (2005), Kandárová et al. (2009, 2005)
70	Decanoic acid (capric acid)	334–48-5	Cat. 2		4.26–6.82	3.1	6.1–17.6			43.6	Han et al. (2021), Kandárová et al. (2009), OECD (2011a), Tornier et al. (2010)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
71	3,3-Dimethylpentane	562–49-2	Cat. 2		93.09–109.24		72.3–90.8			88.3	Han et al. (2021), Kandárová et al. (2009), OECD (2011a)
72	2-Methylbutyric acid	600–07-7	Cat. 1	5.1–8.7		0.6–1.1					Alepee et al. (2019, 2015b)
73	Di-n-propyl disulphide	629–19-6	Cat. 2	5.4–81	87.1–91.2	0.95–2.62	61.9–104.2	18.3–96.8		2.45–62.51	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh et al. (2009), OECD (2011b)
74	Dicaprylyl Ether	629–82-3	Cat. 2	97.2	100.8	36.1	81.7				Sugiyama et al. (2018)
75	Triethoxysilane	998–30-1	Cat. 2	27.3	24.2						Forlini et al. (2025)
76	Sodium chloroacetate	3926–62-3	Cat. 2	69.1							Alepee et al. (2015a)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
77	1-Methyl-3-phenylpiperazine	5271–27-2	Cat. 2	9.47–17.4	5.45–55.07	9.22	12.1–20.5	1.9–91.4	4.5	-1.86–30.84	European Commission (2016), Kandárová et al. (2009), Katoh and Hata (2011), OECD (2021), OECD (2025a, b, c), Spielmann et al. (2007), Tornier et al. (2010)
79	D-Limonene	5989–27-5	Cat. 2	5.9–33.14	8.1–23.6	0.7–1.6					Cotovio et al. (2005), Kandárová et al. (2004, 2006), Zuang et al. (2002)
81	Dimethoxydiphenylsilane	6843–66-9	Cat. 2	109.1	6.3						Forlini et al. (2025)
82	Benzenethiol, 5-(1,1-dimethylethyl)-2-methyl	7340–90-1	Cat. 2	12.74–87.75	8.4–14.58	8.46	22.2–37.2	2.4–24.0		-11.12–34.8	European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Katoh and Hata (2011), Spielmann et al. (2007), Tornier et al. (2010)
84	Copper (II) sulphate	7758–98-7	Cat. 2	0–22.1							Alepee et al. (2016)
85	Clove leaf oil	8000–34-8	Cat. 2			0.8–1.3					Fiume et al. (2016)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
86	Thyme oil, red	8007–46-3	Cat. 2			0.8–1.3					Fiume et al. (2016)
88	Mixture of isomers: 1-(2-isopropylphenyl)-1-phenylethane (CAS# 191,044–60-7) 1-(3-isopropylphenyl)-1-phenylethane (CAS# 191,044–59-4)1-(4-isopropylphenyl)-1-phenylethane (CAS# 2320–06-1)	52,783–21-8	Cat. 2	66.65–85.67	107.62–116.45						Spielmann et al. (2007)
90	Piroctone olamine	68,890–66-4	Cat. 2	4.8–7.3						1.7	Alepee et al. (2016), Han et al. (2021)
91	Sodium lauryl ether sulphate (30% aq)	68,891–38-3	Cat. 2	25.9							Alepee et al. (2015a)
92	Tallow polypropylene polyamine	68,911–79-5	Cat. 1	203.43	16.4–28.5	12.8–60.2					Cotovio et al. (2005), Fiume et al. (2016), Kandárová et al. (2005)
93	Mixture of isomers: ethyl exo-tricyclo[5.2.1.0(2,6)] decane-endo-2-carboxylate/ethyl endo-tricyclo[5.2.1.0(2,6)] decane-exo-2-carboxylate	80,657–64-3	Cat. 2	7.78–12.05	18.27–58.78						Spielmann et al. (2007)
94	2-Chroromethyl-3,5-dimethy l-4-methoxypyridine HCl	86,604–75-3	Cat. 2	3.92–5.67	5.14–9.33	0.7–0.8	10.7–16.4	-0.1–1.1		1.85–8.03	Alepee et al. (2015b), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh and Hata (2011), Spielmann et al. (2007)
95	(±) Trans-3,3-dimethyl-5-(2,2,3-trimethyl-cyclopent-3-en-1-yl)-pent-4-en-2-ol	107,898–54-4	Cat. 2	8.83–11.71	7.8–14.07						Spielmann et al. (2007)
96	4-Methyl-8-methylenetricyclo[3.3.1.1(3,7)] decan-2-ol	122,760–84-3	Cat. 2	4.54–12.36	8.25–17.12						Spielmann et al. (2007)
97	4-Methyl-8-methylenetricyclo [3.3.1.1(3,7)]dec-2-yl acetate	122,760–85-4	Cat. 2	13.33–31.5	12.73–19.52	1.7					Fiume et al. (2016), Spielmann et al. (2007)
98	(E,E)-3,7,11-trimethyldodeca-1,4,6,10-tetraen-3-ol	125,474–34-2	Cat. 2	4.76–21.91	7.76–40.27						Spielmann et al. (2007)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
99	2-Isopropyl-2-isobutyl-1,3-dimethoxypropane	129,228–21-3	Cat. 2	21.48–89.54	65.02–103.58						Spielmann et al. (2007)
101	Linalool	78–70-6	Cat. 3	14.67	3.62–79.3		7.9–25.4				Kandárová et al. (2009, 2005), OECD (2011a)
102	2-Bromobutane	78–76-2	Cat. 3			1.5–1.9					Alepee et al. (2015b)
103	Eugenol	97–53-0	Cat. 3	5.1–9.57	4.88–12.6	0.0–0.1	18.5–32.2				Alepee et al. (2019, 2015b), Cotovio et al. (2005), Kandárová et al. (2009, 2005), OECD (2011a)
104	2-Methyl-4-phenyl-2-butanol	103–05-9	Cat. 3	6.5–16.6	9.7–66.3	6.6–78					Kandárová et al. (2004, 2006), Zuang et al. (2002)
105	dl-Citronellol	106–22-9	Cat. 3	5.9–11.4	8.0–12.3	0.3–1.0					Kandárová et al. (2004, 2006), Zuang et al. (2002)
106	10-Undecenoic acid	112–38-9	Cat. 3	6.0–31.05	5.3–17.5	2.7–14.1					Alepee et al. (2019), Cotovio et al. (2005), Kandárová et al. (2004, 2006), Zuang et al. (2002)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
107	Linalyl acetate	115–95-7	Cat. 3	43.15–103.8	40.4–100.5	1.25	86.8–101.2				Alepee et al. (2019), Cotovio et al. (2005), Kandárová et al. (2009), OECD (2011a), Tornier et al. (2010)
108	Tri-isobutyl phosphate	126–71-6	Cat. 3	4.4–8.3	6.4–44.63	1.18–2.4					Alepee et al. (2019), Alépée et al. (2010), Kandárová et al. (2009), Spielmann et al. (2007), Tornier et al. (2010)
109	Allyl heptanoate	142–19-8	Cat. 3	88.6–112.8	87.4–112	55.42–101.7	45.9–118.9				Alepee et al. (2019, 2015b), Alépée et al. (2010), Kandárová et al. (2009), Katoh et al. (2009)
110	Oleyl Alcohol	143–28-2	Cat. 3	93.7		106.4	104.1				Sugiyama et al. (2018)
111	Sodium carbonate (50% aq.)	497–19-8	Cat. 3	83.1–91.3		92.6–99.2					Alepee et al. (2019, 2015b)
112	Dimethyl disulphide	624–92-0	Cat. 3	11.6–53.4	5.6–19.5	2.4–6.4					Kandárová et al. (2004, 2006), Zuang et al. (2002)
113	Trimethoxy (propyl)silane	1067–25-0	Cat. 3	95.4	4.1						Forlini et al. (2025)
114	1,9-Decadiene	1647–16-1	Cat. 3	17.2–20.0		1.3–2.3					Alepee et al. (2019, 2015b)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
115	(Ethylenediaminepropyl) Trimethoxysilane	1760–24-3	Cat. 3	93.1							Alepee et al. (2015a)
116	2-Ethoxyethyl methacrylate	2370–63-0	Cat. 3	65.3–116.3	4.8–104.8		29.3–74.4				Alepee et al. (2019), Kandárová et al. (2009, 2005), OECD (2011a)
117	33% Sodium undecylenate (in aqueous solution)	3398–33-2	Cat. 3	9.4–28		0.8–1.7					Alepee et al. (2019, 2015b)
118	2-Ethylhexyl-4-methoxycinnamate (octinoxate)	5466–77-3	Cat. 3			92.4–106.1				106.5	Fiume et al. (2016), Han et al. (2021)
119	Heptyl butyrate	5870–93-9	Cat. 3	94.7–121.6	76.7–109.3	91.5–104.89	97.2–118.8	94.0–111.0	79.0	70.89–109.76	Alepee et al. (2019, (2015b), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh et al. (2009), OECD (2011b), OECD (2025a, b, c)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
120	Hexyl salicylate	6259–76-3	Cat. 3	85.0–112.9	84.8–108.4	86.31–106.37	88.4–119.5	82.0–103.4	89.1	68.88–107.89	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh and Hata (2011), OECD (2025a, b, c)
122	Mixture of: 2-Methyl-4-(2′,2′,3′-trimethyl-3′-cyclopenten-1′-yl)-4-penten-1-ol 56% (1′R,2R) & 40%(1′R,2S) isomer	14,864–90-6	Cat. 3	6.2–11.09	8.53–14.47		9.4–13.4				OECD (2011a), Spielmann et al. (2007)
123	Mixture of: 5-Exo-decylbicyclo[2.2.1]hept-2-ene; 5-endodecylbicyclo[2.2.1]hept-2-ene	22,094–85-5	Cat. 3	103.47–107.00	101.24–109.22						Spielmann et al. (2007)
124	1-[4-(2-Dimethylamino ethoxy) phenyl]-2-phenylbutan-1-one	68,047–07-4	Cat. 3	4.84–13.44	43.27–68.26						Spielmann et al. (2007)
125	2-Methyl-3-[(1,7,7-trimethylbicyclo[2.2.1]hept-2-yl) oxy]-1-propanol, bornyl	128,119–70-0	Cat. 3	4.44–10.81	7.9–11.43						Spielmann et al. (2007)
126	Mixture of isomers: 1-(1,1-dimethylpropyl)-4-eth-oxycis-cyclohexane;/1-(1,1-dimethylpropyl)-4-eth-oxytrans-cyclohexane	181,258–87-7cis,181,258–89-9trans	Cat. 3	31.32–53.55	100.12–103.72						Spielmann et al. (2007)
127	Piperonyl butoxide	51–03-6	NC	147.1							Alepee et al. (2015a)
128	Glycerol	56–81-5	NC		94.95–107.27		98.1–125.7				Kandárová et al. (2009), OECD (2011a)
129	Propylene glycol	57–55-6	NC							84.5	Han et al. (2021)
130	2-Phenylethanol (phenylethyl alcohol)	60–12-8	NC	63.1–104.4	45.68–92.91	5.64				0.3	Alepee et al. (2019), Han et al. (2021), Spielmann et al. (2007), Tornier et al. (2010)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
132	Isopropanol	67–63-0	NC	71.1–103.2	76.8–109.2	88.15–105.92	74.2–94.5	95.0–113.3	90.7	65.27–99.49	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Kandárová et al. (2005), Katoh et al. (2009), OECD (2011b), OECD (2025a, b, c)
133	Salicylic acid	69–72-7	NC	7–22.5							Alepee et al. (2016)
134	Triethoxy (vinyl)silane	78–08-0	NC	104.9	106.7						Forlini et al. (2025)
135	Diethyl phthalate	84–66-2	NC	56.9–115.3	83.4–102.15	66.99–100.46	55.4–104.3	91.4–114.6		74.06–103.28	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Hn et al. (2020), Kandárová et al. (2009), Katoh and Hata (2011), Spielmann et al. (2007)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
136	Naphthalene acetic acid	86–87-3	NC	75.5–98.7	98.32–109.5	91.95–103.9	92.7–115.3	82.0–114.4	94.6	61.14–109.52	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh and Hata (2011), OECD (2011b), OECD (2025a, b, c), Spielmann et al. (2007)
138	2,4-Xylidine (2,4-dimethylaniline)	95–68-1	NC	10.3–21.4	-5.4–14.1	0–1.3					Kandárová et al. (2004, 2006), Zuang et al. (2002)
139	Benzyl alcohol	100–51-6	NC	72.46	4.2–98.6		5.6–17.8				Kandárová et al. (2009, 2005), OECD (2011a)
140	1-(4-Chlorophenyl)-3-(3,4-dichlorophenyl) urea (Triclocarban)	101–20-2	NC	111.2						102.3	Alepee et al. (2015a), Han et al. (2021)
141	Phenethyl bromide (2-bromoethy benzene)	103–63-9	NC	5.5–7.2		0.9–7.8					Alepee et al. (2019, 2015b)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
142	Hydroxycitronellal	107–75-5	NC	16.1–83.3	7.8–100.3	60.3–116.0	19.8–32.3				Alepee et al. (2019), Kandárová et al. (2004, 2006), OECD (2011a), Zuang et al. (2002)
143	Isopropyl myristate	110–27-0	NC	102.8–127.41	92.6–106.2	94.4–122.7	97.2–107.9			101.9	Cotovio et al. (2005), Han et al. (2021), Kandárová et al. (2005, 2006), OECD (2011a), Sugiyama et al. (2018)
144	Squalane	111–01-3	NC	95.8	96.1	105.3	92.5				Sugiyama et al. (2018)
145	Ethoxydiglycol (2-(2-ethoxyethoxy)ethanol)	111–90-0	NC	108.1–111.8							Alepee et al. (2016)
146	Triethylene glycol	112–27-6	NC	90.7–116.9	94.95–95.29	96.5–101.42				76.6	Alepee et al. (2019), Fiume et al. (2016), Han et al. (2021), Spielmann et al. (2007), Tornier et al. (2010)
147	Dodecanol (Lauryl Alcohol)	112–53-8	NC	7.1	93.9	2.4	93.1				Sugiyama et al. (2018)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
148	Methyl stearate	112–61-8	NC	85.6–109.6	93.7–105.9	91.32–112.5	78.3–113.1	84.0–117.0	101.7	72.5–112.19	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Kandárová et al. (2005), Katoh and Hata (2011), OECD (2011b), OECD (2025a, b, c)
149	Erucamide	112–84-5	NC	97.8–106.2	86.7–124.0	95.1–107.7	90.0–102.5			103.1	Alepee et al. (2019, 2015b), Han et al. (2021), Kandárová et al. (2005), OECD (2011a)
150	Benzyl salicylate	118–58-1	NC	93.7–121.37	87.0–103.2	86.5–113.3	93.6–99.9			98.5	Alepee et al. (2019, 2015b), Cotovio et al. (2005), Han et al. (2021), Kandárová et al. (2009, 2005), OECD (2011a)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
151	4,4'-Methylenebis(2,6-di-tert-butylphenol)	118–82-1	NC	71.6–111.2	79.6–128	86.5–115.1	100.0–103.9			97.3	Alepee et al. (2019, 2015b), Han et al. (2021), Kandárová et al. (2005, 2006), OECD (2011a), Zuang et al. (2002)
152	Benzyl benzoate	120–51-4	NC	110.03	89.4–109.8	84.3–104.1	99.6–105.7			98.3	Alepee et al. (2015b), Han et al. (2021), Kandárová et al. (2009, 2005), OECD (2011a)
153	1-Chloro-3-nitrobenzene (3-chloronitrobenzene)	121–73-3	NC	13.4–89.95	11.6–103.6	79.9–113.9	95.2–104.3			87.7	Han et al. (2021), Kandárová et al. (2009, 2006), OECD (2011a), Zuang et al. (2002)
154	2-Hydroxy-4-methoxybenzophenone (Benzophenone-3)	131–57-7	NC	95.9–108.4						99.3	Alepee et al. (2016)
155	Benzyl acetate	140–11-4	NC	21.67	73–113.6		11.3–37.4				Kandárová et al. (2009, 2005), OECD (2011a)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
156	Isopropyl palmitate	142–91-6	NC	93.9–104.32	87.0–122.3	102.1	101.8–115.9			106.0	Alepee et al. (2019), Cotovio et al. (2005), Han et al. (2021), Kandárová et al. (2009, 2005), OECD (2011a), Sugiyama et al. (2018)
157	Dodecanoic acid (lauric acid)	143–07-7	NC	102.5–127.6	14.5–105.7	56.6–119.2	88.8–110.0			84.8	Alepee et al. (2019), Han et al. (2021), Kandárová et al. (2009, 2005, 2006), OECD (2011a), Sugiyama et al. (2018)
158	Sodium bicarbonate	144–55-8	NC	92–104.2	82.2–105.5	78.0–113.3	99.6–100.3			91.0	Alepee et al. (2019, 2015b), Han et al. (2021), Kandárová et al. (2009, 2005, 2006), OECD (2011a)
160	3-Chloro-4-fluoronitrobenzene	350–30-1	NC	3.9–88.4	53.19–101.67	101.59				80.7	Alepee et al. (2019), Han et al. (2021), Spielmann et al. (2007), Tornier et al. (2010))
161	Sodium benzoate	532–32-1	NC	90.6–106.5							Alepee et al. (2016)
162	Cyclotetrasiloxane	556–67-2	NC	101.5	96.3	97.5	96.6				Sugiyama et al. (2018)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
163	4-Amino-4H-1,2,4-triazole (4-amino-1,2,4-triazole)	584–13-4	NC	71.0–113	43.5–99.1	80.2–118.5	97.4–101.0			89.7	Alepee et al. (2019), Han et al. (2021), Kandárová et al. (2004, 2006), OECD (2011a), Zuang et al. (2002)
164	n-Butyl propionate	590–01-2	NC	31.52–104	12.75–132.5		23.1–45.9				Alepee et al. (2019), Cotovio et al. (2005), Kandárová et al. (2009, 2005), OECD (2011a)
166	1,5-Hexadiene	592–42-7	NC		93.59–113.17		85.6–95.6				Kandárová et al. (2009), OECD (2011a)
168	2-Methylresorcinol	608–25-3	NC	46.7–124.4							Alepee et al. (2015a, 2016)
169	Triethylsilane	617–86-7	NC	8.6	109.3						Forlini et al. (2025)
170	1,6-Dibromohexane	629–03-8	NC	8.6–29.99	9.5–15.1						Alepee et al. (2019), Cotovio et al. (2005), Kandárová et al. (2005)
)171	cis-Cyclooctene	931–87-3	NC	8.0–44.3	9.1–97.3	0.6–0.9					Kandárová et al. (2004, 2006), Zuang et al. (2002)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
172	3,3'-Dithiodipropionic acid	1119–62-6	NC	60.6–109	66.0–112.1	82.9–117.1	89.9–100.0			96.3	Alepee et al. (2015b), Han et al. (2021), Kandárová et al. (2004, 2006), OECD (2011a), Zuang et al. (2002)
173	Zinc oxide	1314–13-2	NC	105.5–111.8							Alepee et al. (2016)
174	6-hydroxyindole	2380–86-1	NC	70.6–90.7							Alepee et al. (2016)
175	[3-(2,3-Epoxypropoxy)propyl]trimethoxysilane	2530–83-8	NC	11.5	26.4						Forlini et al. (2025)
176	3-Trimethoxysilylpropyl methacrylate	2530–85-0	NC	115.4	102.8						Forlini et al. (2025)
177	3-(Chloropropyl)trimethoxysilane (Silane A-1430)	2530–87-2	NC	36–94.6	61.72–98.62	80.68				3.3	Alepee et al. (2019), Han et al. (2021), Spielmann et al. (2007), Tornier et al. (2010)
178	Benzyl acetone	2550–26-7	NC	22.1–31.8		0.6–11.1					Alepee et al. (2019, 2015b), Spielmann et al. (2007)
179	Cyclohexadecanone	2550–52-9	NC	112.38–121.85	7.85–113.61		98.2–106.6			103.9	Han et al. (2021), OECD (2011a), Spielmann et al. (2007)
180	3,4-Dimethyl-1H-pyrazole	2820–37-3	NC	2.6–31.3	7.58–14.07						Spielmann et al. (2007)
182	[3-(2,3-Epoxypropoxy)propyl]diethoxymethylsilane	2897–60-1	NC	111.7	3.4						Forlini et al. (2025)
182	Trimethoxyphenylsilane	2996–92-1	NC	6.6	8.9						Forlini et al. (2025)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
183	1,1,3,3-Tetramethyldisiloxane	3277–26-7	NC	87	105.9						Forlini et al. (2025)
184	4-Methylthio-benzaldehyde	3446–89-7	NC	6.3–67.6	5.5–96.12	2.65–5.15	12.2–25.7	45–102.7		-0.85–27.42	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh et al. (2009), OECD (2011b), Spielmann et al. (2007)
185	2-Phenylhexanenitrile	3508–98-3	Cat. 3	86.67–116.18	73.87–82.14					76.0	Han et al. (2021), Spielmann et al. (2007)
186	4-Acetoxy-2,5-dimethyl-3(2H)-furanone	4166–20-5	NC	98.18–114.34	80.3–91.2	102.31				4.6	Han et al. (2021), OECD (2010), Spielmann et al. (2007), Tornier et al. (2010)
187	Tetrabromophenol blue	4430–25-5	NC	108.3–112						89.9	Alepee et al. (2016), Han et al. (2021)
188	Octyldodecanol (50%)	5333–42-6	NC	93.5		90.5	94.9				Sugiyama et al. (2018)
189	3-Diethylaminopropionitrile	5351–04-2	NC	27.66–63.07	23.41–76.8	1.61					Spielmann et al. (2007), Tornier et al. (2010)
190	Ammonium nitrate	6484–52-2	NC	104.0							Alepee et al. (2015a)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
191	1-Bromo-4-chlorobutane	6940–78-9	NC	4.02–5.95	5.2–88.85	0.85–10.54	11.0–44.8	60.1–99.4		2.9–23.57	Alépée et al. (2010), European Commission (2016), Han et al. (2020), Kandárová et al. (2009), Katoh et al. (2009), OECD (2010); OECD (2011b), Spielmann et al. (2007)
192	2S-(2-furyl)-5R-hydroxy-4R-(1R,2-dihydroxy) ethyl-6Shydroxymethyl-1,3-dioxane	7089–59-0	NC	99.72–111.79	97.66–106.13						Spielmann et al. (2007)
194	Diisopropyl sebacate	7491–02-3	NC	91.1		91	104.6			93.8	Han et al. (2021), Sugiyama et al. (2018)
195	Allyl phenoxyacetate	7493–74-5	NC	71.7–119.7	73.4–107.5	47.28–95.00	55.4–99.2	0.9–20.7		52.8–92.5	Alepee et al. (2019), Alépée et al. (2010), European Commission (2016), Han et al. (2020), Hwang et al. (2020), Kandárová et al. (2009), Katoh et al. (2009), OECD (2011b)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
196	Sodium bisulphite	7631–90-5	NC	83–107.97	46.24–100.2	79.1–99.7	11.1–74.7				Alepee et al. (2019, 2015b), Cotovio et al. (2005), Kandárová et al. (2009, 2005), OECD (2011a)
197	Sodium bisulphate	7681–38-1	NC			0.5–0.6					Alepee et al. (2015b)
198	Sodium dithionite	7775–14-6	NC	107.97							Cotovio et al. (2005)
199	Mineral Oil	8042–47-5	NC	101.1	99.5	95.8	84.5				Sugiyama et al. (2018)
200	10% Xanthan gum (in D.W.)	11,138–66-2	NC	101.3		98.4	94			99.6	Han et al. (2021), Sugiyama et al. (2018)
202	Talc	14,807–96-6	NC	99.1	97.4	103.8	107.6				Sugiyama et al. (2018)
203	Amino-2-hydroxy-3-Pyridine	16,867–03-1	NC	93.2–121.6							Alepee et al. (2016)
204	2,6-Dimethyl-4-nitrobenzeneamine	16,947–63-0	NC	100.53–107.13	97.21–107.38	114.35					Spielmann et al. (2007), Tornier et al. (2010)
205	2,4,11,13-tetraazatetradecanediimidamide,N,N-bis(4-chlorophenyl)-3,12-diimino-, di-D-gluconate (20% aq)	18,472–51-0	NC	2.8							Alepee et al. (2015a)
206	Ethylhexyl Stearate	22,047–49-0	NC	95.0	81.3	91.2	99.5				Sugiyama et al. (2018)
207	Octyldodecyl Myristate (10%)	22,766–83-2	NC	102.5		96.9					Sugiyama et al. (2018)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
208	Dipropylene glycol	25,265–71-8	NC	81.5–111.0	74.9–116.4	82.99–106.10	92.9–113.3			94.9	Alepee et al. (2019), Alépée et al. (2010), Han et al. (2021), Kandárová et al. (2009, 2005), OECD (2011a, 2011b)
209	Polyethylene glycol 400 (PEG-400)	25,322–68-3	NC	101.4	90.86–112.32	93.4	98.2			96.1	Han et al. (2021), Kandárová et al. (2009), Sugiyama et al. (2018)
210	2-Ethylhexyl-4-aminobenzoate	26,218–04-2	NC	90.87–111.44	91.94–107.34					103.1	Han et al. (2021), Spielmann et al. (2007)
211	Basic brown 16	26,381–41-9	NC	72.3–96.1							Alepee et al. (2016)
212	Phenylbenzimidazole sulfonic acid	27,503–81-7	NC	93.3–108.9							Alepee et al. (2016)
213	Ethylhexyl Palmitate	29,806–73-3	NC	95.7	111.9	91.0	97.4				Sugiyama et al. (2018)
214	Dipropylene glycol monobutyl ether	29,911–28-2	NC	85.0–113.9	98.19–102.54	97.37				13.9	Alepee et al. (2019), Han et al. (2021), Spielmann et al. (2007), Tornier et al. (2010)
215	2-(Formylamino)-3-thiophenecarboxylic acid	43,028–69-9	NC	89.02–95.82	97.8–105.7	107.99					Spielmann et al. (2007), Tornier et al. (2010)
216	3-Mercaptohexanol	51,755–83-0	NC	15.52–62.63	43.02–63.39	2.27–39.6	30.1				Spielmann et al. (2007), Sugiyama et al. (2018), Tornier et al. (2010)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
217	2-Methyl-5-hydroxyethylaminophenol	55,302–96-0	NC	102.4–115.4							Alepee et al. (2016)
218	Propiconazole	60,207–90-1	NC	5.1							Alepee et al. (2015a)
219	Trisodium 4-oxido-2-oxo-5-(oxoacetyl)-2,5-dihydrofuran-3-ylphosphate	66,170–10-3	NC	94.4							Alepee et al. (2015a)
220	2,4-Diaminophenoxyethylaminophenol	66,422–95-5	NC	90.7–105.7							Alepee et al. (2016)
221	Ethyl cis-4-[4-[[2-(2,4-dichlorophenyl)-2-(1H-imidazol-1-ylmethyl)-1,3-dioxolan-4-yl] methoxy] phenyl]piperazine-1-carboxylate	67,914–69-6	NC	113.74–118.51	82.95–95.08						Spielmann et al. (2007)
222	Basic yellow 87	68,259–00-7	NC	92–109.3							Alepee et al. (2016)
223	Basic red 76	68,391–30-0	NC	105.4–111.5							Alepee et al. (2016)
224	Basic yellow 57	68,391–31-1	NC	109.8–124.7							Alepee et al. (2016)
225	Basic Brown 17	68,391–32-2	NC	95.5–100.8							Alepee et al. (2016)
226	Mixture of: diethyl cis-1,4-Cyclohexanedicarboxylate; diethyl trans-1,4-cyclohexanedicarboxylate	72,903–27-6	NC	67.94–99.92	97.75–106.67						Spielmann et al. (2007)
227	HC Yellow n°9	86,419–69-4	NC	72.6–97.4							Alepee et al. (2016)
228	1-Hydroxyethyl-4.5-diamino pyrazole sulphate	155,601–30-2	NC	94.5–103.2							Alepee et al. (2016)
229	Disodium 2,2′-(1,4-phenylene)bis-1H-benzimidazole-4,6-disulfonic acid or monosulfonic acid, monosulfonate or disulfonate	180,898–37-7	NC	93.4–100.3	92.87–106.21						Spielmann et al. (2007)
230	Dimethicodiethylbenzalmalonate(Parsol SLX)	207,574–74-1	NC			77.6–108					Fiume et al. (2016)
231	Capryl-isostearate	209,802–43-7	NC	95.95–102.33	99.46–108.02					94.1	Han et al. (2021), Spielmann et al. (2007)
232	Mixture of isomers: 1-(spiro[4.5]dec-7-en-7-yl) pent-4-en-1-one/1-(spiro[4.5]dec-6-en-7-yl) pent-4-en-1-one	224,031–70-3	NC	40.40–66.43	51.21–87.65						Spielmann et al. (2007)

Table 3 (continued)

ID	Substance	CAS RN	Reference category	Viability (%)							Reference
				EpiSkin	EpiDerm	SkinEthic	LabCyte	epiCS	Skin	KeraSkin	
233	Diethylamino hydroxybenzoyl hexyl benzoate	302,776–68-7	NC	115.6–121.7							Alepee et al. (2016)
234	Propyl (2S)-2-(1,1-dimethylpropoxy)-propanoate	319,002–92-1	NC	58.06–85.07	86.88–99.34						Spielmann et al. (2007)

Abbreviations: ID, internal substance identification number; CAS RN, Chemical Abstracts Service Registry Number; EpiSkin, EpiSkin™ (VRM); EpiDerm, EpiDerm™ SIT (EPI-200) (VRM); SkinEthic, SkinEthic™ RHE; LabCyte, LabCyte EPI-MODEL24 SIT; epiCS, epiCS®; Skin, Skin+®; KeraSkin, KeraSkin™ SIT; RhE, reconstructed human epidermis; Reference category, harmonized category assigned in the ChemSkin 2.0 database. Viability values are presented as minimum–maximum (%) ranges when multiple independent experimental results were available for the same substance. For substances with multiple published records, all retrieved viability values were considered in defining the displayed range; to avoid redundancy, only references corresponding to the extreme values are indicated in the table

between primary viability records (Table 3) and derived performance metrics (Table 5), the database structure supports transparent interpretation of RhE model behavior under borderline conditions. These findings confirm the detection of Category 2 irritants while highlighting the regulatory and biological constraints inherent to binary irritation classification systems.

Physical and chemical property–dependent model performance

Physical state-dependent model performance

To investigate whether experimental properties influence model performance, predictive accuracy was stratified by the physical state of the substances (liquid, solid, and gel) (Table 6).

For TG 431, liquid substances generally demonstrated higher predictive accuracy than solids across several RhE models. Chi-square analysis revealed statistically significant differences in accuracy between physical states for SkinEthic™ RHE ($p=0.038$) and LabCyte EPI-MODEL24 SCT ($p=0.017$). The relatively lower accuracy observed for solid substances in these models may reflect differences in exposure behavior on the RhE surface, while liquids distribute uniformly across the system, solids may exhibit variability in dissolution of surface contact during the exposure period, potentially affecting the consistency of corrosive action.

In TG 439, several models—including SkinEthic™ RHE, LabCyte EPI-MODEL24 SIT and KeraSkin™ SIT, showed higher accuracy for solid substances than for liquids. However, no statistically significant differences were observed across the seven TG 439 models (all $p>0.05$), suggesting that physical state was not a primary determinant of predictive variability for irritation hazard.

Gel substances showed high accuracy in most models, but these results should be interpreted cautiously due to the limited number of substances in this state, which restricts the stability of the observed performance.

Chemical category-dependent model performance

The predictive performance was further evaluated across predefined chemical categories to identify potential mechanistic biases (Table 7).

For TG 431 corrosion models, consistently high accuracy was observed for strong acids/bases and reactive organics. These classes typically involve rapid cytotoxic or protein-reactive mechanisms that produce pronounced reductions in cell viability. In contrast, lower and more variable accuracy values were observed for surfactants and certain inorganic

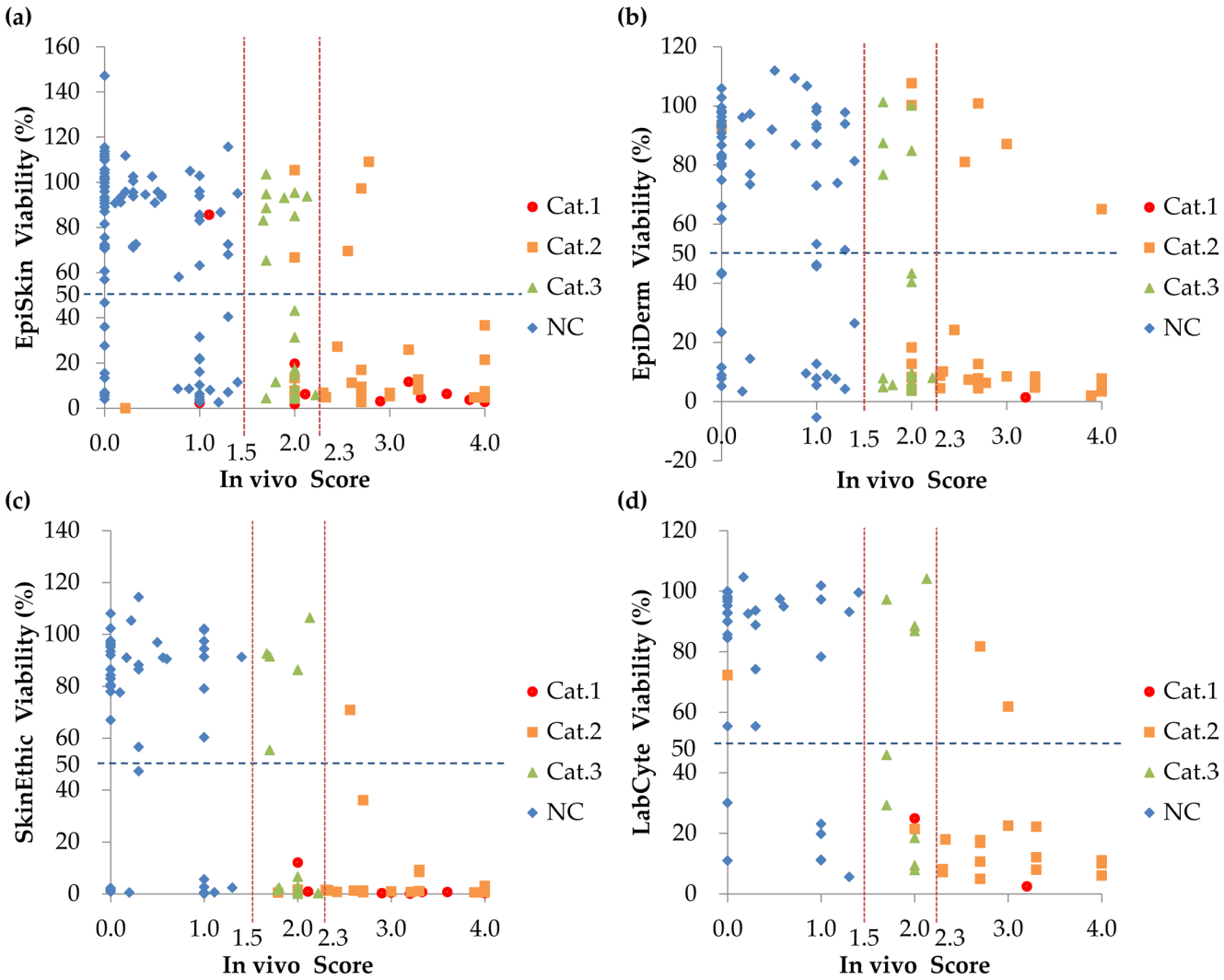


Fig. 8 Relationship between in vivo irritation scores and RhE cell viability values across four OECD TG 439 RhE models. **a** EpiSkin™ (VRM), **b** EpiDerm™ SIT (EPI-200), **c** SkinEthic™ RHE, **d** LabCyte EPI-MODEL24 SIT. Horizontal dashed lines indicate the 50% viability threshold used for irritation classification according to OECD TG 439, while vertical lines indicate in vivo irritation classification boundaries

compounds. These groups may exert membrane-disruptive mechanisms or indirect tissue-damaging effects that are less consistently captured by the viability-based endpoint.

Specifically, surfactants may induce membrane-disruptive effects through lipid solubilization and the perturbation of epidermal barrier components, often resulting in progressive rather than abrupt cytotoxic responses (Jackson et al. 2014; Welss et al. 2004). This observation is consistent with previous studies indicating that intrinsic chemical reactivity and structural features are key determinants of skin irritation and corrosion potential (Hulzebos et al. 2005). Collectively, these findings suggest that while RhE models are robust for identifying direct-acting toxicants, their predictive performance for mechanistically complex classes like surfactants is more sensitive to the structural integrity and barrier properties of the specific reconstructed tissue system.

Chi-square analysis indicated statistically significant category-dependent differences for two TG 431 models (SkinEthic™ RHE, $p=0.047$; LabCyte EPI-MODEL24 SCT, $p=0.027$), supporting the interpretation that intrinsic chemical reactivity contributes to variability in predictive performance across chemical classes.

A similar but more heterogeneous pattern was observed for TG 439 irritation models. While strong acids/bases and neutral organic solvents generally demonstrated relatively high concordance, greater variability was observed for reactive organics and surfactants. Unlike corrosion testing, irritation classification relies on a single 50% cell viability threshold. Consequently, substances inducing moderate or borderline cytotoxicity are more susceptible to classification shifts resulting from experimental variability, particularly within chemically diverse classes (Tornier et al. 2010).

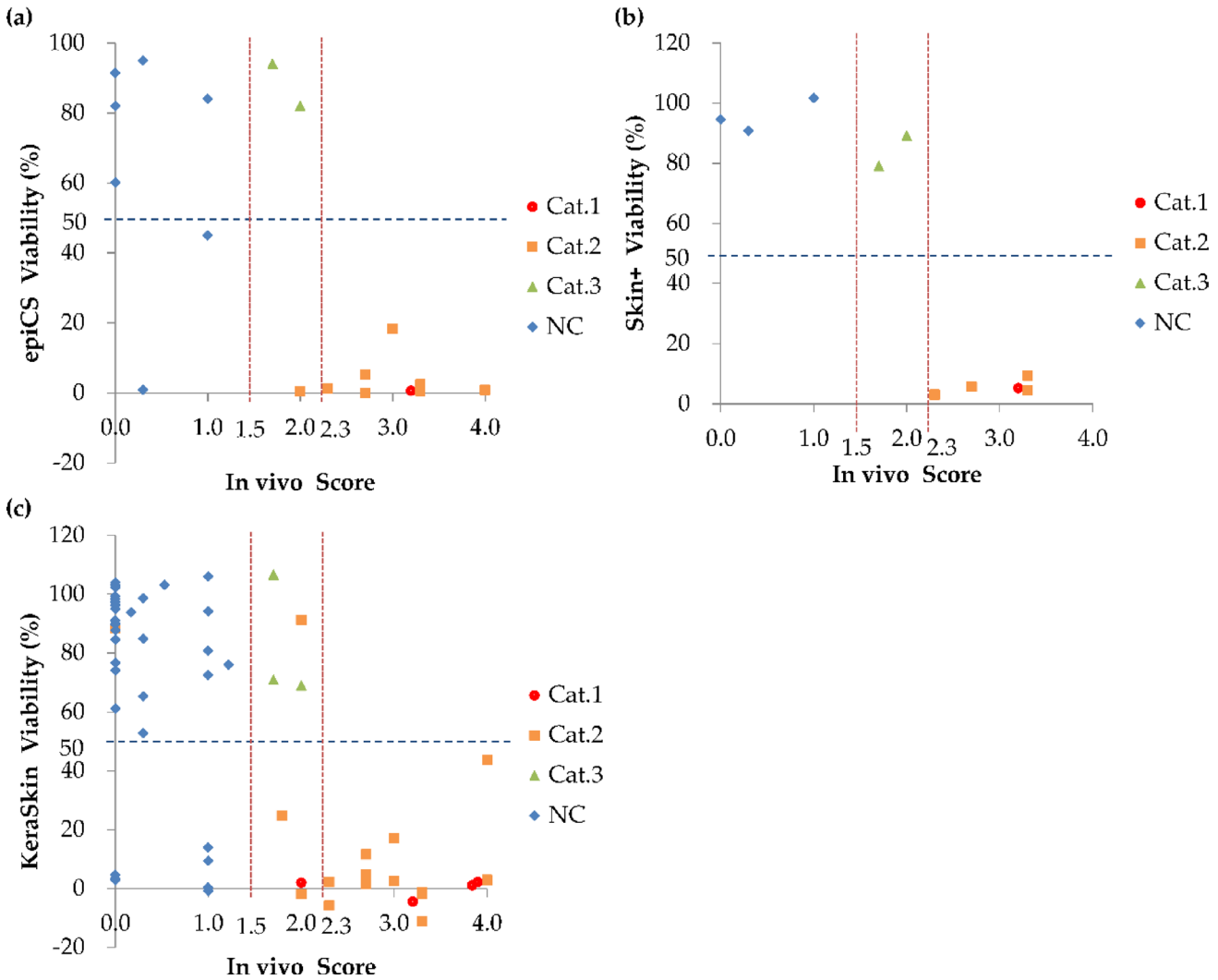


Fig. 9 Relationship between in vivo irritation scores and RhE cell viability values across additional three OECD TG 439 RhE models. **a** epiCS®, **b** Skin + ®, **c** KeraSkin™ SIT. The data representation and decision thresholds are consistent with those described in Fig. 8

Table 4 Correlation between in vivo irritation scores and RhE cell viability across OECD TG 439 models

RhE model	N	Spearman's ρ	*p*-value
EpiSkin™ (VRM)	165	− 0.567	< 0.001**
EpiDerm™ SIT	117	− 0.523	< 0.001**
SkinEthic™ RHE	104	− 0.624	< 0.001**
LabCyte EPI-MODEL24 SIT	65	− 0.630	< 0.001**
epiCS®	20	− 0.611	< 0.001**
Skin + ®	10	− 0.778	0.008**
KeraSkin™ SIT	60	− 0.653	< 0.001**

Spearman's ρ: rank correlation coefficient between in vivo irritation score and RhE cell viability. ** Correlation is significant at the 0.01 level (two-tailed)

Category-dependent variability appeared more pronounced for TG 439 models compared with TG 431 models. This difference is consistent with the structural contrast between the two guidelines: TG 431 incorporates exposure-time-dependent decision criteria that help stabilize categorical outcomes, whereas TG 439 relies on a binary viability cutoff that compresses the irritation severity gradient. As a result, irritation predictions appear more sensitive to subtle mechanistic differences and threshold-adjacent responses.

Collectively, these findings indicate that predictive performance is not uniformly distributed across chemical classes, highlighting the importance of chemical-category stratification when interpreting RhE model outcomes. By integrating structural categorization with harmonized reference categories, the ChemSkin 2.0 database enables mechanistically informed interpretation of model behavior across the corrosion–irritation continuum.

Nevertheless, interpretation of category-dependent differences should be approached cautiously for chemical groups represented by limited numbers of substances. In such cases, the restricted sample size may reduce statistical

Table 5 Predictive performance of OECD TG 439 RhE models relative to the reference category

OECD TG 439 RhE Model	N	Sensitivity (%)	Specificity (%)	Accuracy (%)	Kappa
EpiSkin™ (VRM)	195	87.0	66.9	74.9	0.506
EpiDerm™ SIT (EPI-200) (VRM)	119	80.6	60.2	66.4	0.339
SkinEthic™ RHE	131	98.5	65.6	82.4	0.646
LabCyte EPI-MODEL24 SIT	67	85.0	74.5	77.6	0.527
epiCS®	20	100.0	77.8	90.0	0.794
Skin+®	10	100.0	100.0	100.0	1.000
KeraSkin™ SIT	62	91.3	82.1	85.5	0.702

N, number of substances evaluated per model; Sensitivity, proportion of reference Category 2 irritants correctly predicted; Specificity, proportion of reference non-irritant substances (including Category 3 consolidated as NC under EU CLP implementation) correctly predicted; Accuracy, overall proportion of correct predictions; Kappa, Cohen's kappa coefficient calculated relative to the reference category. Predicted classifications were derived using the minimum reported viability value relative to the 50% threshold defined in OECD TG 439

stability and affect expected cell frequencies in contingency analyses, such as Pearson's chi-square test. Consequently, patterns observed within smaller chemical categories should be interpreted as indicative trends rather than definitive conclusions regarding model performance.

Conclusions

The ChemSkin 2.0 database establishes a harmonized, severity-oriented reference framework for skin corrosion and irritation by integrating regulatory classifications and experimental evidence across the full hazard spectrum. By consolidating 234 substances under a unified reference category system aligned with UN GHS criteria and traceable to EU CLP classifications, the database provides a transparent and regulatory-consistent foundation for evaluating in vitro model performance.

The database structure links substance-level reference categories with model-specific RhE viability records, enabling direct comparison between regulatory classifications, in vivo observations, and in vitro responses. Preservation of minimum–maximum viability ranges maintains traceability to primary experimental data and supports precautionary interpretation for substances located near regulatory decision thresholds.

Performance analyses demonstrated that OECD TG 431 models reliably identify corrosive substances and show substantial agreement with the reference categories. Severity-resolved analysis further indicated partial preservation of the hierarchical gradient within Category 1 (1A–1C), suggesting that RhE systems capture both binary corrosion outcomes and aspects of potency differentiation.

In contrast, TG 439 irritation models showed moderate to substantial agreement, with variability primarily driven by specificity differences. This reflects the structural characteristics of irritation classification, which relies on a single viability threshold and consolidates mild irritation (Category 3) into the non-classified (NC) group under EU CLP implementation. These findings illustrate how regulatory decision architectures influence the interpretation of validation metrics and apparent model performance.

Stratified analyses revealed that model performance varied according to both physical state and chemical category, highlighting the influence of mechanistic and experimental factors on RhE-based predictions. For example, strong acids/bases and reactive organics showed consistently high concordance, while greater variability was observed for surfactants and chemically heterogeneous classes. This highlights the necessity of considering chemical structure and mode of action when interpreting in vitro test outcomes.

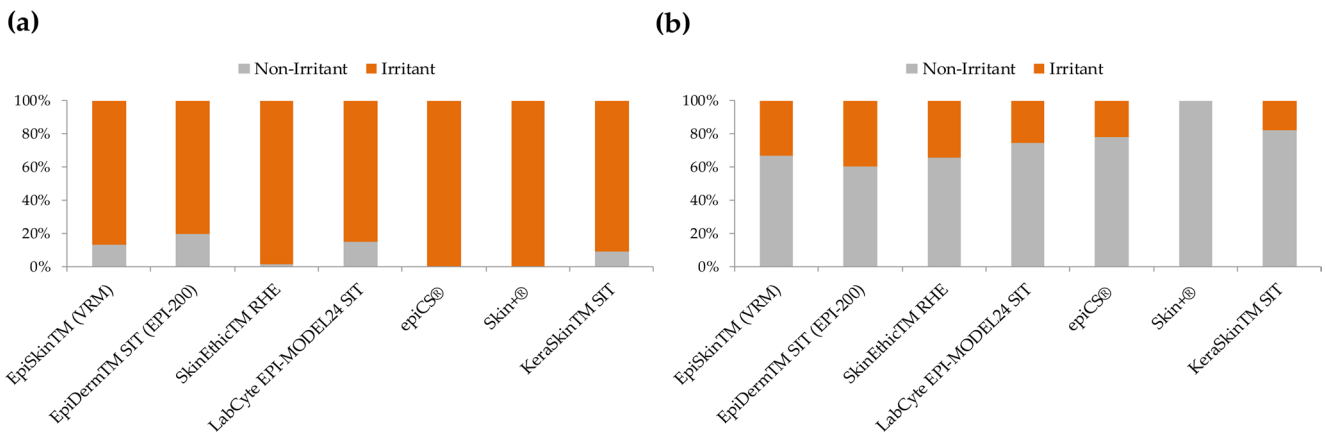


Fig. 10 Predictive classifications of OECD TG 439 RhE models relative to the reference category. **a** Predicted outcomes for substances classified as irritant (Category 2); **b** Predicted outcomes for substances classified as non-irritant (including Category 3 consolidated as NC under EU CLP implementation)

Table 6 Accuracy of OECD TG 431 and TG 439 RhE models stratified by physical state

Physical State	OECD TG 431					OECD TG 439					
	EpiSkin™ (SM)	EpiDerm™ SCT (EPI-200)	Ski-nEthic™ RHE	epiCS®	LabCyte EPI-MODEL24 SCT	EpiSkin™ (VRM)	EpiDerm™ SIT (EPI-200) (VRM)	Ski-nEthic™ RHE	LabCyte EPI-MODEL24 SIT	epiCS®	Skin+®	Ker-aSkin™ SIT
(Total N)	(106)	(85)	(116)	(72)	(73)	(195)	(119)	(131)	(67)	(20)	(10)	(62)
Liquid	86.7	83.9	88.5	87.0	89.3	72.8	65.2	81.1	72.0	87.5	100.0	81.0
Solid	72.7	73.9	72.4	83.3	64.7	80.7	69.2	87.0	93.8	100.0	100.0	94.4
Gel	100.0					50.0	100.0	100.0	100.0			100.0
Total Accuracy	84.0	81.2	84.5	86.1	83.6	74.9	66.4	82.4	77.6	90.0		85.5
Chi-square *p*-value (<0.05)	0.255	0.297	0.038*	0.694	0.017*	0.368	0.720	0.645	0.166	0.456		0.333

Values represent prediction accuracy expressed as percentages (%). Pearson's chi-square test was used to evaluate differences in prediction accuracy between physical states; *$p<0.05$. Values should be interpreted cautiously for categories represented by a small number of substances

Overall, ChemSkin 2.0 provides a robust and regulatory-aligned platform for evaluating RhE model performance across the corrosion–irritation continuum. By integrating harmonized reference categories, severity stratification, and preserved experimental variability, the database supports improved interpretation of in vitro test results and facilitates the development of Integrated Approaches to Testing and Assessment (IATA).

Future research should utilize the preserved viability ranges to investigate threshold-adjacent substances and regulatory "grey zones." Integration with complementary New Approach Methodologies (NAMs) and structure-informed analyses may enhance predictive robustness and support more consistent, severity-oriented hazard assessment strategies.

Table 7 Accuracy of OECD TG 431 and TG 439 RhE models stratified by chemical category

Chemical Category	OECD TG 431					OECD TG 439					
	EpiSkin™ (SM)	EpiDerm™ SCT (EPI-200)	Ski-nEthic™ RHE	epiCS®	LabCyte EPI-MODEL24 SCT	EpiSkin™ (VRM)	EpiDerm™ SIT (EPI-200) (VRM)	Ski-nEthic™ RHE	LabCyte EPI-MODEL24 SIT	epiCS®	Skin + ®	Ker-aSkin™ SIT
(Total N)	(106)	(85)	(116)	(72)	(73)	(195)	(119)	(131)	(67)	(20)	(10)	(62)
Strong acids/bases	86.2	86.7	86.7	88.0	87.5	87.0	50.0	95.8	100.0	100.0	100.0	100.0
Surfactants	50.0	75.0	80.0	50.0	50.0	62.5	100.0	80.0	100.0			100.0
Reactive organics	85.7	82.4	85.7	86.7	83.9	69.2	59.1	83.0	70.0	80.0	100.0	88.5
Neutral organic solvents	88.5	70.0	87.0	80.0	100.0	72.7	70.5	76.2	81.1	100.0	100.0	82.6
Aromatic sulfonates / dyes / salts			0.0			100.0	100.0	100.0	100.0	100.0		100.0
Inorganic compounds	60.0	71.4	85.7	100.0	40.0	100.0	66.7	75.0	66.7			50.0
Others (rare case)						50.0	0.0	50.0	50.0	100.0		50.0
Total Accuracy	84.0	81.2	84.5	86.1	83.6	74.9	66.4	82.4	77.6	90.0		85.5
Chi-square *p*-value (<0.05)	0.188	0.740	0.047*	0.496	0.027*	0.101	0.320	0.428	0.751	0.695		0.432

Values represent prediction accuracy expressed as percentages. Pearson's chi-square test was used to assess category-dependent differences in predictive accuracy; *$p<0.05$. Values should be interpreted cautiously for categories represented by a small number of substances

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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